Begränsad delning

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Computing Students' Appraisals of and Coping with Risks from Artificial Intelligence

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Sammanfattning

Artificiell intelligens integreras i allt högre grad i utbildning, arbetsliv, offentliga tjänster och vardagsliv, vilket skapar både möjligheter och risker. Syftet med denna studie var att undersöka hur IT-studenter bedömer och prioriterar AI-relaterade risker, samt hur de resonerar kring möjliga sätt att hantera dessa risker. Studien genomfördes med hjälp av sju semistrukturerade intervjuer med svenska studenter på kandidat- och masternivå inom områden som mjukvaruteknik, datavetenskap, systemvetenskap, tillämpad AI, AI och automation samt interaktionsdesign. Analysen vägleddes av Technology Threat Avoidance Theory och en taxonomi över AI-risker, samtidigt som ytterligare teman tilläts framträda ur intervjumaterialet. Studien visar att deltagarna främst oroade sig för omedelbara och vardagsnära risker, särskilt överdriven tillit till AI, förlorad kompetens, minskat kritiskt tänkande, okritisk användning och AI:s påverkan på utbildning och professionell kompetens. Risker kopplade till mer avancerade AI-system, såsom förlorad kontroll eller existentiell risk, uppfattades generellt som avlägsna, osannolika och mindre akuta. Resultaten visar även att deltagarnas förståelse av vad AI är påverkade vilka risker de uppfattade som relevanta. Föreslagna hanteringsstrategier handlade främst om utbildning, ansvarsfull användning, granskning av AI-genererade svar och begränsad tillit till AI-system. Samtidigt placerade deltagarna ofta ansvaret för bredare riskhantering hos myndigheter, företag eller andra institutioner, men uttryckte begränsad tillit till dessa aktörer. Studien drar slutsatsen att IT-studenter är medvetna om många AI-relaterade risker, men att både utbildningsinsatser och strukturella åtgärder behövs för att stärka framtida hantering av AI-risker.

Nyckelord: artificiell intelligens, AI-riskperception, TTAT, IT-studenter, riskhantering

Abstract

Artificial intelligence is increasingly integrated into education, work, public services, and everyday life, creating both opportunities and risks. The aim of this study was to examine how computing students appraise and prioritise AI-related risks, and how they reason about possible responses to such risks. The study was conducted through seven semi-structured interviews with Swedish computing students from bachelor's and master's programmes in fields such as software engineering, computer science, systems science, applied AI, AI and automation, and interaction design. The analysis was guided by Technology Threat Avoidance Theory and an AI risk taxonomy, while also allowing additional themes to emerge from the interview material. The study shows that participants were most concerned with immediate and everyday risks, especially overreliance, deskilling, reduced critical thinking, uncritical trust, and the effects of AI on education and professional competence. Risks related to advanced AI systems, such as loss of control or existential risk, were generally perceived as distant, unlikely, and less urgent. The findings also show that participants' understanding of what AI is shaped which risks they considered relevant. Coping responses focused mainly on education, responsible use, verification of AI outputs, and limiting reliance on AI systems. At the same time, participants often placed responsibility for broader risk mitigation on governments, authorities, or companies, while expressing limited trust in these actors. The study concludes that computing students are aware of many AI-related risks, but that educational and structural interventions are needed to strengthen future AI risk mitigation.

Keywords: artificial intelligence, AI risk perception, TTAT, computing students, risk management

Preface

I would like to express my sincere gratitude to my supervisor, Thomas Persson, for his high-quality feedback, thoughtful guidance, and level of engagement throughout this thesis process. His support has far exceeded my expectations and has been highly valuable for the development of this work.

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Terminology

Acronyms

AI	Artificial Intelligence
TTAT	Technology Threat Avoidance Theory
HCI	Human-computer interaction
LLM	Large Language Model

1. Introduction

The impact of artificial intelligence (AI) on our lives is increasing as it is being integrated widely in our society in areas ranging from education to finance (Maslej et al., 2025; Said et al., 2023; Zhang and Dafoe, 2019). This is reflected in academia where AI papers made up 41.8% of total publications in computer science in 2023, almost doubling since 2013 (Maslej et al., 2025). While there are many potential benefits with AI ranging from personalised learning in education (Matos et al., 2025) to reduced costs and improved diagnostics in healthcare (El Arab and Al Moosa, 2025), there are also many potential risks (Floridi et al., 2018; Russell and Norvig, 2022, p. 1038; Said et al., 2023; Zhang and Dafoe, 2019). The type of risks considered in the academic literature are wide, ranging from contemporary issues like misinformation and job-loss (Weichert et al., 2025), to hypothetical future threats like existential risk from misaligned AI (McLean et al., 2023).

To facilitate the mitigation of the risks from AI, research is needed to understand AI risk perception among the general public, and particularly among groups capable of influencing the adoption and development of AI systems (Said et al., 2023; Weichert et al., 2025). The developers of AI systems have the potential to affect the risks from these systems by withholding their work, in their technical decisions and by raising concerns in the development process (Griffin, Green and Welie, 2024). Some also argue that developers could be ethically responsible for controlling and managing risks associated with the AI systems they create, or at least to mitigate these risks (Douglas, Lacey and Howard, 2025). At the same time, studying professional developers directly can be challenging due to limited access and organisational constraints (Duke, 2022), making computing students a relevant and more accessible group for examining how future developers may perceive and respond to AI-related risks (Weichert et al., 2025). Bingley et al. (2023) even go so far as using students as proxies for developers in their study on human-centred artificial intelligence.

Weichert et al. (2025) identify computing students as future developers and deployers of AI systems, as well as potential AI policymakers, and reason that their attitudes towards AI ethics are therefore a critical area of study;

furthermore, they identify this area as underexplored. While Weichert et al. (2025) explore the attitudes of computing students towards some AI risks, their focus is broader and their questionnaire largely leaves out some categories of risks. Ghotbi and Ho (2021) have explored which risks, framed as ethical issues, related to AI students in higher education find most important and found increasing unemployment to be the main concern, while malicious AI and AI control of society was of much lower concern. These studies by Weichert et al. (2025) and Ghotbi and Ho (2021) suggest that students may prioritise and appraise AI risks, although the underlying reasoning behind such prioritisation and appraisal remains insufficiently explored.

Existing research suggests that developers and computing students recognise AI-related risks, but their responses vary. Some acknowledge ethical responsibility yet shift it towards users or organisations, while others intervene by raising concerns or adjusting their career choices (Griffin, Green and Welie, 2024). Similarly, students often perceive themselves as capable of mitigating risks, but may be reluctant to engage in broader governance or policy-related efforts (Weichert et al., 2025).

These findings indicate that computing students may recognise AI risks while selectively engaging in different forms of mitigation. However, the reasoning underlying both how different risks are appraised and prioritised, and how coping responses are appraised, remains insufficiently understood. Addressing this gap is important for AI governance, safety, and education, as computing students form one group from which future developers, deployers, and technically informed decision-makers may emerge. Furthermore, understanding their reasoning may help inform the design of computing education by providing insight into how students interpret and engage with AI-related risks. This is further underscored by calls for stronger integration of ethical reasoning in computing education, highlighting the importance of understanding how students engage with AI-related risks (Griffin, Green and Welie, 2024).

1.1 Purpose

The purpose of this study is to develop an in-depth understanding of how computing students appraise and prioritise different types of AI-related risks and how they reason about possible coping responses to such risks. This is important as these students form one group from which future AI developers, deployers, and technically informed decision-makers may emerge. Their perceptions of AI-related risks may therefore provide insight into how such risks could be understood and addressed in practice.

1.1.1 Research questions

RQ1. How do computing students appraise and prioritise different types of AI-related risks?

RQ2. How do the students reason about possible coping responses to these risks?

2. Extended background

2.1 The general public and AI risk

Research on the general public suggests that perceptions of AI-related risks are shaped by factors such as knowledge, age, education, and technical experience. Both Said et al. (2023) and Zhang and Dafoe (2019) found that greater familiarity with AI and technical competence are associated with more positive attitudes toward AI development and lower perceived concern regarding AI-related risks and governance challenges. Younger individuals similarly tended to perceive lower levels of risk (Said et al., 2023; Zhang and Dafoe, 2019). At the same time, Said et al. (2023) found that higher levels of AI knowledge were associated with "risk blindness," where individuals underestimated AI-related risks in balanced scenarios. The authors suggest this may stem from overconfidence in one's understanding of AI systems and one's ability to manage associated risks. They also found that unwarranted confidence in AI knowledge was linked to more extreme perceptions of both risks and benefits (Said et al., 2023).

Taken together, these findings suggest that greater familiarity with AI does not necessarily lead to heightened concern about its risks. Instead, knowledge and experience may be associated with increased confidence and reduced perceived risk, as well as more polarised attitudes. This highlights the importance of further investigating AI risk perception, particularly among groups with the capacity to influence the development and deployment of AI systems (Said et al., 2023).

2.2 Developers and AI risk

Developers play a central role in both the creation and mitigation of AI-related risks through their technical decisions, their ability to raise concerns, and their influence over how systems are designed and deployed (Griffin, Green and Welie, 2024; Douglas, Lacey and Howard, 2025). As a result, some scholars argue that developers carry ethical responsibility for the societal consequences of the systems they create (Douglas, Lacey and Howard, 2025). This has led to growing interest in how developers themselves understand AI-related risks, responsibility, and ethical agency.

Research suggests that developers often hold ambivalent and sometimes contradictory views regarding their responsibility for AI risks. Studies of AI developers indicate that they frequently describe technology and technical work as morally neutral, which can contribute to shifting responsibility toward users, organisations, or society at large (Griffin, Green and Welie, 2024). Similarly, Duke (2022) shows that healthcare AI developers tend to acknowledge risks while simultaneously downplaying their significance by framing them as temporary technical challenges that future innovation will solve. In both studies, developers often minimise their own role in contributing to these risks, despite recognising that AI systems may have significant societal consequences.

At the same time, developers do demonstrate forms of ethical agency. Griffin, Green and Welie (2024) found that some developers actively raise ethical concerns during development processes or reconsider their employment when projects conflict with their personal values. However, both Griffin, Green and Welie (2024) and Duke (2022) highlight how organisational and economic incentive structures can discourage such

actions by prioritising innovation, efficiency, and financial outcomes over ethical reflection and risk mitigation.

Taken together, these findings suggest that developers may recognise AI-related risks while simultaneously distancing themselves from responsibility or downplaying the significance of these risks. At the same time, they retain the capacity to act on ethical concerns under certain conditions. This highlights the importance of understanding how individuals in technical roles interpret and respond to AI-related risks, particularly given their influence over the development and deployment of such systems.

2.3 Computing students and AI risk

Computing students have been identified as an important group in research on AI ethics and risk, as they form one group from which future developers, deployers, and potentially policymakers of AI systems may emerge (Weichert et al., 2025). Their attitudes towards AI-related risks and ethical issues are therefore considered highly relevant for understanding how such risks may be addressed in practice. At the same time, this area of research remains relatively underexplored (Weichert et al., 2025).

Empirical studies suggest that computing students tend to prioritise certain types of AI-related risks over others. Ghotbi and Ho (2021) found that students in higher education most frequently identified unemployment due to automation as the primary ethical concern, while risks such as malicious AI or loss of societal control received very little attention. This indicates that students may focus more on immediate and tangible risks compared to more abstract or long-term concerns.

Similarly, Weichert et al. (2025) found that computing students tend to centre on current risks with their biggest concerns being deepfakes, misinformation, data privacy and job-loss. More speculative or long-term risks, such as existential threats from AI, were largely absent from predefined response categories and were not raised by the participants. However, qualitative responses in their study suggest that students may

still consider a broader range of issues, such as environmental impacts, when given the opportunity to elaborate beyond structured survey formats.

In terms of responses to AI-related risks, Weichert et al. (2025) report that a majority of the computing students perceived themselves as capable of mitigating potential harms associated with AI systems. At the same time, relatively few students expressed interest in pursuing roles related to AI policy or regulation, despite recognising the importance of such roles (Weichert et al., 2025). This suggests a potential gap between ability and willingness to engage in certain forms of risk mitigation.

Taken together, existing research indicates that computing students are aware of AI-related risks and often express confidence in their ability to address them, while simultaneously prioritising certain categories of risk over others. However, the underlying reasoning behind how these risks are prioritised, and how different forms of response are evaluated, remains insufficiently understood. This highlights the need for more in-depth qualitative research into how computing students interpret and respond to AI-related risks.

3. Theory

3.1 AI systems

The Organisation for Economic Co-operation and Development defines an AI system as:

> "An AI system is a machine-based system that, for explicit or implicit objectives, infers, from the input it receives, how to generate outputs such as predictions, content, recommendations, or decisions that can influence physical or virtual environments. Different AI systems vary in their levels of autonomy and adaptiveness after deployment" (OECD, 2024).

This thesis adopts the OECD (2024) definition because it captures AI as a broad functional category rather than as one specific technology. Under

this definition, AI systems are machine-based systems that take some form of input, process it in relation to explicit or implicit objectives, and generate outputs that may influence digital or physical environments. These outputs can include predictions, classifications, recommendations, decisions, or generated content.

The OECD (2024) definition builds on a conceptual view of AI systems presented by Russell and Norvig (2022), where an AI system can be understood as consisting of three main components: sensors, operational logic, and actuators. Sensors collect or receive data from the environment. The operational logic processes this input in relation to the system's objectives. Actuators then produce outputs or actions that affect the environment, either physically or virtually (OECD, 2019; Russell and Norvig, 2022). In a physical robot, sensors may include cameras or other measurement devices, while actuators may include motors or mechanical components. In a digital AI system, inputs may instead consist of text, images, user behaviour, or database records, while outputs may include recommendations, generated text, rankings, classifications, or decisions.

3.2 AI risk

In this study, AI risks are understood broadly as potential negative consequences arising from the development, deployment, use, or societal integration of AI systems. This includes both consequences that are currently occurring, as well as those that may arise in the future. The study does not aim to evaluate the objective validity or probability of such risks, but rather focuses on how they are perceived and reasoned about by computing students. As such, different risks may be interpreted and prioritised differently depending on individuals' knowledge, experiences, and expectations.

While participants' own interpretations are central, existing research is used to provide a conceptual overview of the range of risks discussed in the literature. Prior work has identified a wide variety of AI-related risks, spanning technical, social, and ethical domains. To structure this space, this study draws on the taxonomy proposed by Slattery et al. (2026), which

groups AI risks into seven overarching categories. While the study adopts an inclusive approach to AI risks and allows participants' own interpretations to guide which potential harms they consider relevant, the existing literature on AI risks still acts as a guide for understanding why some risks may or may not be salient to the students. The seven domains are summarised in table 1 and a more thorough review of the risk domains can be found in Appendix B.

Category	Description	References
Discrimination and toxicity	AI systems can reproduce and amplify societal biases through training data and design choices, leading to discriminatory outcomes (e.g., in hiring or law enforcement) and harmful stereotypical representations; they may also expose users to toxic content and perform unevenly across demographic groups, resulting in exclusion and reduced trust.	Slattery et al., 2026; Wirtz et al., 2022; Kamila and Jasrotia, 2025; Shelby et al., 2023; Paes et al., 2023; Hagendorff, 2024; Weidinger et al., 2022; Wirtz et al., 2020.
Privacy and security	AI systems can expose, infer, or leak sensitive information through data collection, memorisation, or interaction patterns, enabling both direct and indirect privacy breaches as well as unintended disclosure of personal or proprietary data; they are also vulnerable to security threats such as adversarial attacks, system manipulation, and exploitation of weaknesses in infrastructure, integrations, and design.	Slattery et al., 2026; Hagendorff, 2024; Shelby et al., 2023; Cunha and Estima, 2023.
Misinformation	AI systems can generate and amplify false, misleading, or low-quality information due to limitations in data, design, and optimisation processes, potentially leading to inaccurate beliefs and harmful decisions in high-stakes contexts; at scale, they can also contribute to the fragmentation of information environments through personalised content, reinforcing filter bubbles, reducing trust in institutions, and increasing societal polarisation.	Slattery et al., 2026; Cunha and Estima, 2023; Hagendorff, 2024; Shelby et al., 2023; Weidinger et al., 2022; Nah et al., 2023; Zhang et al., 2024.
Malicious actors and misuse	AI systems can be intentionally exploited to deceive, manipulate, or cause harm through applications such as disinformation, surveillance, cyberattacks, weaponisation, and fraud, with generative capabilities lowering barriers to creating convincing and scalable harmful content; this enables more efficient coordination of attacks, targeted manipulation, and large-scale societal impacts, including threats to democratic processes, security, and trust.	Slattery et al., 2026; Cunha and Estima, 2023; Weidinger et al., 2022; Habbal et al., 2024; Wirtz et al., 2022; Bastos and Mercea, 2019; Hagendorff, 2024; Kilian et al., 2023; Teixeira et al., 2022; Shelby et al., 2023; Nah et al., 2023.
Human–computer interaction	AI systems can lead to overreliance, unsafe use, and emotional dependence as users place undue trust in their outputs or attribute human-like understanding to them, increasing the risk of harmful decisions or inappropriate information disclosure;	Slattery et al., 2026; Hagendorff, 2024; Weidinger et al., 2022; Wirtz et al., 2022; Nah et al., 2023; Paes et

	over time, this reliance may erode critical thinking, reduce autonomy, and contribute to feelings of disempowerment as decision-making is increasingly delegated to opaque systems.	al., 2021; Kamila and Jasrotia, 2025; Shelby et al., 2023.
Socioeconomic and environmental harms	AI systems can contribute to structural inequalities by concentrating power, reshaping labour markets, and devaluing human work, potentially leading to job displacement, precarious employment, and unequal distribution of benefits; they may also exacerbate governance challenges and environmental impacts through resource-intensive development, regulatory gaps, and incentives that prioritise rapid deployment over societal well-being.	Slattery et al., 2026; Weidinger et al., 2022; Shelby et al., 2023; Nah et al., 2023; Giarmoleo et al., 2024; Wirtz et al., 2022; Hagendorff, 2024; Paes et al., 2021; Meek et al., 2016; Cunha and Estima, 2023; McLean et al., 2023; Saghiri et al., 2022; Wirtz et al., 2020; Steimers and Schneider, 2022; Stahl and Eke, 2024.
Safety and limitations	As AI systems become more advanced, they may behave unpredictably, fail in critical contexts, or act in ways misaligned with human intentions due to issues such as goal misalignment, limited robustness, or opaque decision-making processes; these risks are compounded by potentially dangerous capabilities and system interactions, which can increase the likelihood of large-scale harm, loss of control, and challenges for oversight and accountability.	Slattery et al., 2026; Hagendorff, 2024; Wirtz et al., 2022; McLean et al., 2023; Saghiri et al., 2022; Teixeira et al., 2022; Nah et al., 2023; Kamila and Jasrotia, 2025; Meek et al., 2016; Paes et al., 2023; Bourget and Chalmers, 2023; Francken et al., 2022; Motwani et al., 2024.

Table 1: Summary of AI risk categories

3.3 Technology Threat Avoidance Theory

Technology Threat Avoidance Theory (TTAT) by Liang and Xue (2009) is used to understand how the computing students relate to different AI risks. Originally the theory was developed from a synthesis of research from the fields of information systems, risk analysis, psychology and health care with the aim to describe user behaviour in relation to malicious IT in the form of malware or spam (Liang and Xue, 2009). TTAT was chosen because of its established position within computer science and information systems research as an empirically tested theoretical lens for understanding perceptions of IT threats (Chen and Liang, 2019; Liang and Xue, 2009; Liang and Xue, 2010; Herath et al., 2014; Young et al., 2016), and because it is well suited for analysing both risk appraisal and coping appraisal.

Liang and Xue (2009) distinguish between malicious IT and virtuous IT: “Malicious IT is defined as systems perceived by users to be repulsive and

to cause negative outcomes, and virtuous IT is defined as systems perceived by users to be attractive and to cause positive outcomes". They avoid using the developers' intentions behind creating a system and only focus on the user perception which might differ due to conflicting interests and differing contexts (Liang and Xue, 2009). Under these definitions the same AI system could be recognised by certain users as malicious IT and as virtuous IT by others.[1] A developer losing their job due to an AI coding agent replacing them can perceive it as malicious, while a consultant developer that sees the AI coding agent as an opportunity to deliver better products to their client can perceive it as virtuous. Avoiding developer intentions also makes TTAT a good fit when it comes to the case of some of the risks in the category of safety and limitations from Slattery (2026), described in section 7 of Appendix B. Since these scenarios are often described as playing out without the malicious intent of the developers, or perhaps without intentions at all in the case of self-improving AI, user perception is clearly a better fit.

The basic claim of TTAT is that a user perceiving an IT threat is motivated to take protective action if they perceive the protective action to be efficacious towards avoiding the threat, if they do not perceive any protective action as efficacious, they will perform emotion-based coping (Liang and Xue, 2009). TTAT includes two parts, a Process Theory and a Variance Theory (Liang and Xue, 2009).

The Process Theory uses a positive feedback loop from cybernetics theory to explain how humans self-regulate behaviour related to malicious IT (Liang and Xue, 2009). After becoming aware of a malicious IT, the person will take the undesired state of being harmed by the malicious IT as an anti-goal (Liang and Xue, 2009). When the distance between their current state and the anti-goal becomes small enough a threat is perceived (Liang and Xue, 2009). Once a threat has been perceived the person will start a coping appraisal that determines which coping-behaviours they perform and then the coping-behaviours are performed until the distance between

[1] The terminology of user might not be a perfect fit in this thesis. In some of the risks considered the affected is not necessarily a user of the AI-system, but they are nonetheless affected by it and have to relate to it.

the current state and the anti-goal is large enough and the perceived threat disappears (Liang and Xue, 2009). The coping behaviours are divided into two categories: problem-focused coping which is oriented towards attempting to objectively reduce the threat through safe-guarding measures; and emotion-focused coping where one changes one's desires or perceptions of reality to reduce the perceived threat, motivated by the negative feelings associated with the threat (Liang and Xue, 2009). Often a combination of both problem-focused and emotion-focused coping behaviours are used to reduce the sense of threat (Liang and Xue, 2009).

The Variance Theory view of TTAT acts as a snapshot of the process from the Process Theory and adds key influencing factors and relations (Liang and Xue, 2009).

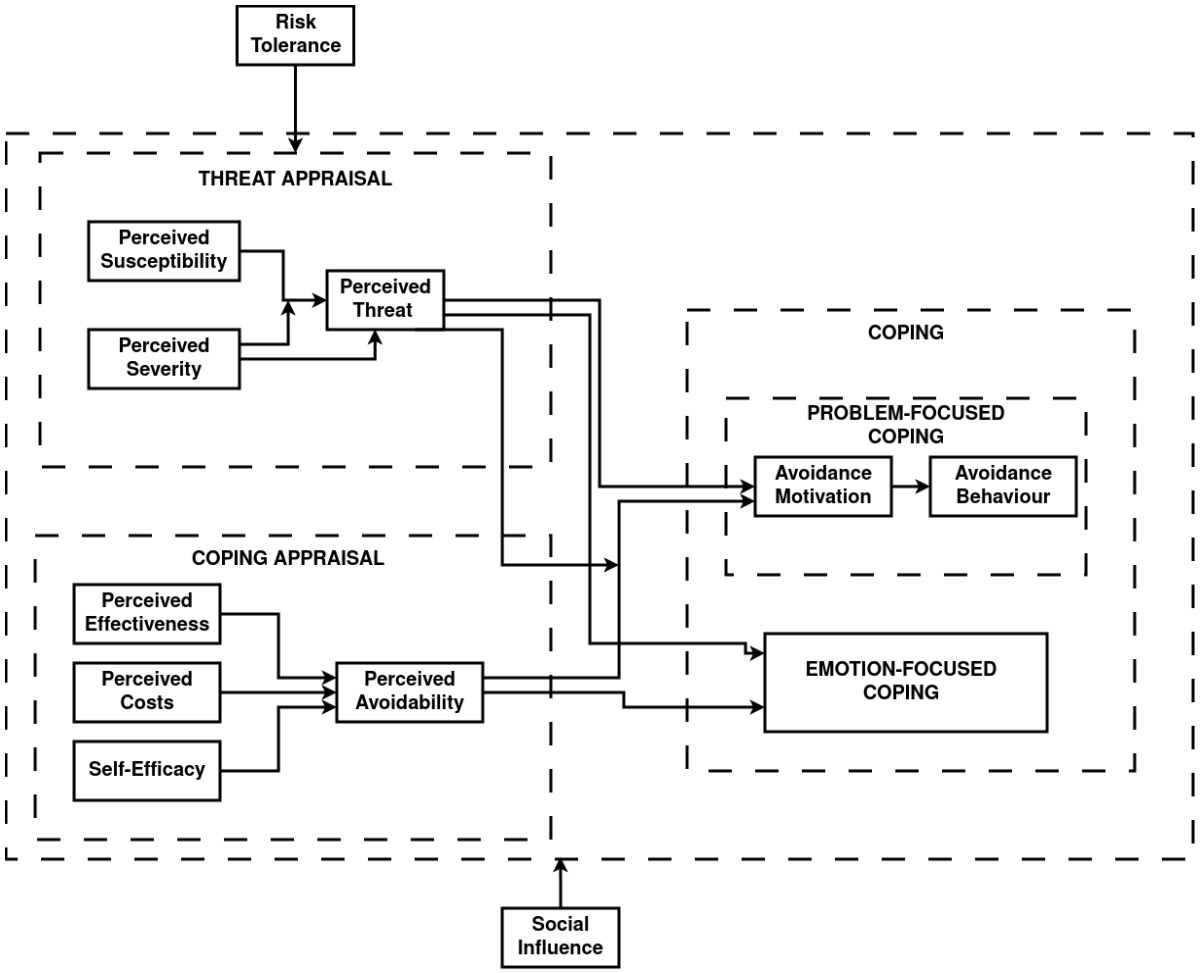


Figure 1. Variance Theory view of TTAT, adapted from Liang and Xue (2009).

3.3.1 Threat Appraisal

As shown in the top left section of figure 1 threat appraisal consists of perceived susceptibility and perceived severity resulting in a perceived threat. This is based on risk analysis research where risk is often seen as the probability of a negative event occurring combined with severity of its consequences (Liang and Xue, 2009) and this is where the AI risks fit into the theory. In this study AI risks will be framed as technology threats that students evaluate through threat appraisal. Perceived severity is determined by how serious the person takes the negative consequences to be, perceived susceptibility is the subjective probability that the negative consequences will occur (Liang and Xue, 2009). The perceived severity and perceived susceptibility interact to determine the threat perception, if one increases the threat will be perceived as more severe and if one decreases the threat will be perceived as less severe. As seen in the top of figure 1 TTAT also includes risk tolerance as a factor that negatively influences threat perception where risk tolerance is defined as the minimum difference that the subject can tolerate between the anti-goal and the subject's current state (Liang and Xue, 2009). Thus, *ceteris paribus*, the higher one's risk tolerance, the less one's perceived threat.

3.3.2 Coping

After the threat appraisal process has resulted in a perceived threat, the individual will begin the coping appraisal process, depicted in the bottom left part of figure 1, where they try to determine the likelihood of a safeguarding measure in helping in the avoidance of the threat, the perceived avoidability (Liang and Xue, 2009). TTAT identifies three key factors influencing perceived avoidability: perceived effectiveness, the subject’s appraisal of the effectiveness of a safeguard to avoid the threat; perceived costs, the subject’s appraisal of the costs of taking a safeguarding measure; and self-efficacy, the subject’s confidence in their ability to take the safeguarding measure. Many safeguards perceived avoidabilities may be compared in the coping appraisal process before the most effective ones are performed as coping behaviours (Liang and Xue, 2009).

After the coping appraisal process is complete the subject will perform either emotion-focused coping, problem-focused coping or a combination. A lower perceived avoidability will result in the subject tending to perform emotion-focused coping and higher perceived avoidability will result in the subject tending towards greater avoidance motivation, and consequently problem-focused avoidance behaviour. The avoidance motivation is affected by both perceived avoidability and perceived threat (Liang and Xue, 2009).

3.3.3 Social influence

TTAT acknowledges the importance of social influence, shown in the bottom of figure 1, on threat avoidance (Liang and Xue, 2009). Social influence occurs in the form of informational influence and normative influence. Informational influence is the fact that information that is utilised in threat and coping appraisal is provided by the social environment. Normative influence acts through three processes: internalisation; compliance; and identification, resulting in individuals tending towards socially desirable behaviours (Liang and Xue, 2009). For example, a student might act in a way that helps in avoiding an AI risk like loss of autonomy due to guidelines from their university without actually experiencing any threat.

4. Method

4.1 Scientific method

This study adopts a qualitative research strategy in order to develop an in-depth understanding of how computing students appraise AI-related risks and reason about possible coping responses. Qualitative methods are well suited for exploring individuals' perceptions, interpretations, and reasoning processes, as they allow for rich, detailed accounts of subjective experiences (Bryman, 2015). This aligns with the aim of the study, which is not to measure the prevalence of specific attitudes, but to understand the students' point of view and their reasoning.

A significant deductive component was used in the study through its grounding in TTAT. There are risks associated with overrelying on theory in qualitative research, such as not seeing the importance of the data or confirmation bias, but a balanced centering of theory is still possible (Collins and Stockton, 2018; Fife and Gossner, 2024). TTAT was used as a lens to facilitate more nuanced interviews and as an initial guide in coding and analysis. A deliberate attempt was made to remain mindful of data that did not correspond with the theory, as recommended by Collins and Stockton (2018) and Fife and Gossner (2024). In the interview process, this meant not constraining the participants' spontaneous inquiry, while in the analysis it meant remaining open to patterns that did not align with TTAT.

4.2 Data collection

4.2.1 Choice of data collection method

Interviews are one of the most commonly used methods within qualitative research (Bryman, 2015, p. 466), and it is a well-established method for data generation within the fields of computer science and information systems (Goldkuhl, 2019). A qualitative semi-structured interview was chosen for its strengths in balancing the fairly clear focus of the research with flexibility and allowing the participants' points of view to be adequately understood (Bryman, 2015, p. 468-469). This was an important decision to ensure that TTAT could be used to successfully set the outline of the interview, while still allowing the participants' point of view, even if it did not overlap with the theory. The flexibility of the semi-structured interview thus assisted in making sure that data that did not align with the theory was not treated as unwanted digressions. Furthermore, the possibility of asking follow-up questions allowed for clarification and elaboration, contributing to a more nuanced understanding of participants' reasoning and interpretations.

4.2.2 Participant selection

A generic purposive sampling approach was used in the study to ensure that the participants could be utilised to satisfactorily answer the research questions (Bryman, 2015, p. 408-414). Participants were purposefully selected based on their enrolment in computing-related bachelor's and master's programmes that may develop their potential to enter professions

where they may influence the development, deployment, or governance of AI systems.

Within this sampling frame, variation was deliberately sought in order to accentuate any common denominators across the entire range and access a maximum variation of perspectives. In particular, variation in exposure to AI-related coursework was considered important, as it may influence perceived competence and understanding of AI-related risks (Said et al., 2023). Additionally, participants were recruited from multiple universities to reduce the likelihood that institution-specific perspectives would dominate the findings, resulting in a geographical spread across Sweden. Table 2 provides an overview of the participants' study backgrounds.

Participants were recruited by contacting programme managers and student associations connected to relevant programmes, asking them to distribute a recruitment letter through their preferred communication channels. In some cases, students were also contacted directly via email or contact forms available on programme websites when a certain programme manager could not be reached or could not distribute the recruitment letter. The recruitment letter can be found in Appendix D.

Participant	Study programme	University	AI-specific programme	AI-specific courses	Semester	Gender	Interview length
1	Bachelor's programme in software engineering	Linnaeus University	No	Yes	Semester 6/6	Male	30 minutes
2	Bachelor Programme in Systems Science - Design of Information Systems	Lund University	No	No	Semester 4/6	Female	48 minutes
3	Bachelor of Science in Computer Science with specialisation in Computer Systems Development	Malmö University	No	No	Semester 4/6	Male	73 minutes
4	Bachelor's Programme in Data Science – Applied AI	Halmstad University	Yes	Yes	Semester 6/6	Male	75 minutes
5	Master of Science in Applied AI - AI and Neuroscience	Luleå University of Technology	Yes	Yes	Semester 8/10	Female	66 minutes
6	Master's programme in AI and Automation	University West	Yes	Yes	Semester 4/4	Male	70 minutes
7	Master of Science in Interaction Design and Technologies	Blekinge Institute of Technology	No	No	Semester 10/10	Female	49 minutes

Table 2. Participant information

4.2.3 Interview guide design

An interview guide, see Appendix A, was created in accordance with the recommendations by Bryman (2015, p. 470-475). The guide was designed to balance openness with analytical focus by structuring the interview into a sequence of thematic sections aligned with the research questions. The structure of the guide reflects the study's focus on both risk appraisal and prioritisation (RQ1), and coping reasoning (RQ2), with distinct sections addressing each of these dimensions.

The initial sections were formulated to elicit participants' general perceptions of AI and to allow them to introduce AI-related risks in their own terms. This open-ended approach was intended to capture which risks were most salient without strong researcher direction. Subsequent sections included questions targeting key aspects of risk and coping appraisal.

To ensure coverage across a broad range of AI-related risks, the interview guide also included a set of prompts based on the established taxonomy of AI risk categories. These prompts were designed to introduce risk areas that might not emerge spontaneously in earlier parts of the interview, thereby enabling exploration of both salient and less immediately articulated risks. From a TTAT perspective, this approach becomes particularly important since the theory suggests that risks may not become salient for several reasons, including low perceived likelihood or severity, but also because of emotion-focused coping, such as avoidance-related responses. Care was also taken not to ascertain any correctness to these risks, using formulations such as "Some people discuss risks such as…". This was intended to minimise the risk of influencing participants' own assessments of severity and likelihood, as well as to reduce the risk of emotional impact as discussed further in section 4.4.4.

4.2.4 Data collection and procedure

Participants received a brief summary of the study prior to the interviews but were not provided with the full interview guide. This was done to encourage an initial focus on participants' own experiences and reflections, rather than introducing theory-driven categories in advance. As the

interview guide included prompts covering a range of risk categories, no prior preparation was deemed necessary.

The interviews were conducted as online video meetings using Microsoft Teams. This format was chosen primarily due to its time and cost efficiency compared to in-person interviews, as well as its ability to capture both verbal and visual communication (Bryman, 2016, p. 492). While online interviews may involve risks related to technical issues, such as poor connection quality potentially affecting communication or transcription (Bryman, 2016, p. 492), no such issues greatly affected the data collection process.

Prior to the interviews, participants were contacted via email and asked to: confirm their consent to participate in the study; confirm their acceptance of Microsoft Teams as the meeting platform; and consent to recording and transcription using the platform's built-in tools. They were informed that the recordings would be deleted after the completion of the study.

The interviews were recorded and automatically transcribed using Microsoft Teams. This allowed the researcher to focus fully on conducting the interviews rather than taking extensive notes, while also improving efficiency in the transcription process. At the beginning of the interview, the researcher verbally presented the purpose of the study, an outline of how the interview would be structured and a reminder that participation was voluntary and that participants could withdraw at any time without providing a reason. Consent to participation, recording, and automatic transcription through the platform's tools was then confirmed. Participants were informed when the recording started.

During the interviews, the guide presented in Appendix A was used as a flexible outline. The overall structure was generally followed, while follow-up questions were used to probe participants' reasoning regarding the perceived severity, likelihood, and relevance of different risks, as well as their reflections on possible responses. The predefined risk categories were introduced only when participants had not already addressed them, allowing for exploration of both spontaneously raised and prompted risks.

Similarly, questions were not repeated if the relevant topics had already been sufficiently covered.

After the automated transcriptions had been checked and corrected by the researcher, a copy was sent to the participant of their transcripts requesting for them to control and get back if anything in the transcripts seemed to be incorrect. None of the participants requested any changes.

4.3 Analysis

The interview material was analysed using a primarily deductive qualitative analysis approach informed by Technological Threat Avoidance Theory (TTAT) (Liang and Xue, 2009) and the AI risk taxonomy proposed by Slattery et al. (2026), but some inductive elements were also present. The goal was not to test or build on the underlying theory but rather to use it to guide analysis of the data. This approach was inspired by deductive qualitative analysis (DQA), where existing theory can be used to structure and guide analysis while still allowing for inductive elements without necessarily involving theory building or refinement (Fife and Gossner, 2024).

The analysis began during transcription, where recordings were checked and adjusted manually in order to ensure accuracy and facilitate initial familiarisation with the material. During this process, preliminary patterns and noteworthy themes were noted, including recurring discussions regarding responsibility attribution and differing conceptions of AI systems.

An initial codebook was then developed deductively from TTAT and the AI risk taxonomy. TTAT-derived codes focused primarily on threat appraisal and coping appraisal processes, including perceived severity, perceived susceptibility, problem-focused coping, emotion-focused coping, self-efficacy, coping effectiveness, coping costs and social influence. In parallel, the seven overarching AI risk categories from Slattery et al. (2026) were given codes used to classify the types of risks discussed by participants. Additional descriptive metadata codes were also included, such as participant ID and whether the risk was probed or brought up

spontaneously, in order to contextualise how participants framed AI-related risks.

The interview transcripts were coded using the qualitative analysis software Taguette through an initial coding pass using the preliminary codebook. Coding was primarily deductive, with segments assigned codes derived from TTAT and the risk taxonomy. However, some inductive coding was also utilised where recurring patterns relevant to the research questions did not fit cleanly within the initial theoretical structure. These inductively developed codes primarily concerned:

- specific forms of coping responses not explicitly specified within TTAT,
- prioritisation and comparative reasoning between risks,
- responsibility attribution,
- broader conceptualisations of AI as either autonomous actors or neutral tools.

These inductive additions were not treated as extensions of TTAT itself, but rather as contextual factors influencing how participants appraised risks and reasoned about coping responses. When necessary, the coded segments would include some additional padding of text for context and one segment was often coded with multiple codes.

Following the initial coding pass, the codebook was revised to clarify code definitions, merge overlapping categories, and incorporate the inductively identified themes. A second coding pass was then conducted using the revised codebook to improve consistency across interviews. The final codebook can be found in Appendix C.

After the coding was completed, coded segments were extracted into a custom Excel workbook designed to support cross-case comparison and thematic overview. This enabled comparison of how participants appraised different categories of AI risk, how they prioritised risks relative to one another, and how different understandings of AI systems shaped threat appraisal and coping reasoning.

4.4 Reflections on quality

4.4.1 Credibility

To strengthen the credibility of the study, several measures were taken throughout the data collection and analysis process. According to Bryman (2015, p. 384), credibility concerns whether the researcher's interpretation of the studied social reality is perceived as plausible and trustworthy. An important aspect of this is ensuring that the findings accurately reflect the participants' perspectives. To support this, a form of respondent validation was used by allowing participants to review the transcripts of their interviews. This enabled participants to verify that their statements had been transcribed correctly and reduced the risk of misunderstandings or misrepresentation of their views. In line with Bryman (2015, p. 385), this strengthens credibility by giving participants the opportunity to confirm that their accounts had been accurately recorded.

The study also relied on established qualitative methods, which is important for credibility (Bryman, 2015, p. 384). Semi-structured interviews were chosen because they are suitable for exploring participants' experiences, reflections, and perceptions in depth. The flexible interview format allowed participants to elaborate on issues they considered important while still ensuring that core themes relevant to the research questions were addressed.

Furthermore, the analysis combined both deductive and inductive approaches. Existing concepts from TTAT and prior research informed parts of the analytical framework, while new themes and patterns were also allowed to emerge from the empirical material itself. This combination can be understood as a form of analytical triangulation, where different analytical perspectives are used to strengthen the robustness and credibility of the findings rather than relying exclusively on either theory-driven or data-driven interpretation (Bryman, 2015, p. 386).

At the same time, some limitations affecting credibility should be acknowledged. The analysis and interpretation of qualitative data inevitably involve researcher subjectivity. Since the coding and thematic

interpretation were conducted by a single researcher, personal assumptions and preunderstandings may have influenced how themes were identified and interpreted. To mitigate this, the analysis aimed to remain closely grounded in the empirical material through continuous reference back to the interview data during coding and interpretation.

4.4.2 Dependability

Dependability concerns whether the research process is conducted in a consistent, transparent, and traceable manner (Bryman, 2015, p. 384-386). According to Shenton (2004), establishing dependability in qualitative research requires clear descriptions of what was planned and carried out in the study, how data collection was conducted, and reflections regarding how well the chosen methods functioned and their effectiveness.

These aspects are addressed throughout the method and discussion chapters of this study. The research design, participant selection, interview procedure, transcription process, and analytical approach are described in detail in order to provide transparency regarding how the study was conducted. The method chapter also refers to appendices containing relevant supporting material, such as the interview guide and codebook.

In addition, documentation was retained throughout the research process, including interview transcripts, coding structures, and theme development. This creates a form of audit trail that makes it possible to follow how interpretations and conclusions were developed from the empirical material.

Reflections regarding the strengths and limitations of the chosen methods, as well as how effectively they functioned in practice, are discussed in section 6.2 of the thesis. This further contributes to dependability by making the research process and its limitations explicit.

At the same time, some limitations should be acknowledged. The analysis was conducted by a single researcher, which may reduce dependability since interpretations and coding decisions were not continuously reviewed by additional researchers. This increases the possibility that certain perspectives or patterns in the material may have been overlooked.

However, this was mitigated by systematically revisiting the empirical material throughout the analysis and grounding interpretations closely in participants' own statements.

4.4.3 Transferability

Transferability in qualitative research concerns the extent to which findings may be applicable in other contexts or settings (Bryman, 2015, p. 384). However, as both Bryman (2015, p. 384) and Shenton (2004) emphasise, qualitative studies are typically context-dependent and based on smaller samples, meaning that the aim is not statistical generalisation. Instead, the responsibility of the researcher is to provide sufficiently detailed contextual descriptions that allow readers themselves to assess whether the findings may be transferable to other contexts.

To support transferability, this study provides descriptions of the research context, participant characteristics, sampling approach, and data collection procedures. Information regarding the participants' educational backgrounds, and experience with AI has been included to provide contextual understanding of the perspectives represented in the material. The method chapter further describes the interview process, sampling criteria, and analytical procedures, while the interview guide is included in the appendix. In line with Shenton (2004), this detailed contextual information enables readers to evaluate whether the findings may be relevant or transferable to similar populations or settings.

The study focuses specifically on university students within computing programmes. As such, the findings may be more transferable to similar groups, particularly younger adults, students, or individuals engaged with digital technologies, than to the broader population.

Nevertheless, the transferability of the findings is limited by the relatively small and context-specific sample. The study does not claim that the identified perceptions or coping strategies are representative of the general population. Instead, the findings should be understood as contextually situated accounts that may contribute to broader understanding when

compared with findings from similar qualitative studies conducted in other contexts.

4.4.4 Ethical and societal considerations

The study was conducted in accordance with established principles of good research practice, grounded in the guidelines outlined by Vetenskapsrådet (2024). In line with these guidelines, participant consent should be informed, voluntary, specific, explicit, revocable, and documented. To ensure this, participants received written information through the recruitment letter describing the purpose of the study, how the interviews would be conducted, that participation was voluntary, and that they could withdraw at any time without explanation or consequences. This information was also reiterated verbally at the beginning of each interview. Consent to participate, as well as consent to recording and transcription using Microsoft Teams' built-in features, was obtained both through email prior to the interview and verbally before recording began. Participants were also informed when the recording started.

Given that the study concerned perceptions of AI-related risks, care was taken to introduce potentially sensitive or alarming topics in a neutral manner in order to minimise unnecessary emotional discomfort. Risks that participants had not mentioned themselves were introduced through formulations such as "some people discuss the risk that…", allowing participants to engage with the topic without the researcher implying that the risk was necessarily valid or severe.

In accordance with the confidentiality and data management recommendations described by Vetenskapsrådet (2024), measures were also taken to reduce the risk of participant identification and to avoid retaining data longer than necessary. The study did not collect unnecessary personally identifiable information, and participants are presented only in generalised terms. Direct quotations were reviewed carefully to avoid revealing identities through contextual details. All collected material was handled confidentially, with recordings and transcripts stored only on university-approved platforms and local storage solutions. Raw interview data was deleted after the completion of the study.

5. Results and analysis

5.1 Appraisal of AI risks

This section covers participants' risk appraisal. The participants' perceptions are mapped onto the categories adapted from Slattery et al. (2026) as a way to organise them. The goal has not been to ensure a perfect fit with Slattery et al. (2026) but rather to use the categories to provide an easier overview. Table 3 provides an overview of the main patterns in participants' threat appraisal across the seven AI risk categories. The table is intended as a summary of dominant tendencies in the material rather than as a quantitative ranking. Each category is discussed in more detail in the following subsections.

Risk category	Most discussed risks	Perceived severity	Perceived susceptibility	Key modifiers of appraisal
Discrimination and toxicity	Bias in recruitment, policing, image generation, and decision systems	Low–medium	High, but mainly societal/group-level	relativised through comparison with human bias and existing discrimination.
Privacy and security	Data leakage, prompting personal data, institutional data systems, cybersecurity	Mixed; moderate overall, higher in sensitive contexts	Mixed	Mental models of AI memory and data handling; comparison with existing data leaks.
Misinformation	Hallucinations, fake images/news, filter bubbles, epistemic erosion	Medium-high	High, especially for less informed users	LLM (Large Language Model) focus; AI outputs perceived as persuasive and human-like; severity relativised by comparison with existing misinformation.

Malicious actors and misuse	Cyberattacks, fraud, deepfakes, manipulation, terrorism	Medium–high	High; often seen as inevitable	relativised as existing harms amplified by AI; scale, speed, and accessibility increased severity.
Human–computer interaction	Overreliance, deskilling, dependency, uncritical trust, educational impacts	High	High, but often projected onto other groups	Everyday visibility; education context; concern for children, students, and less knowledgeable users.
Socioeconomic and environmental harms	Junior developer roles, labour-market disruption, inequality, concentration of benefits	Medium–high	Medium–high for junior and less specialised workers	Personal career relevance; AI competence perceived as protective; systemic/economic framing.
Safety and limitations	Misalignment, loss of control, autonomous decision-making, AI consciousness	Potentially high in hypothetical scenarios	Low, especially in the short-term	Distant time horizon; LLM-focused mental models; low perceived current AI capability.

Table 3. Summary of participants' threat appraisal across AI risk categories.

5.1.1 Discrimination and toxicity

Participants generally framed bias and discrimination in AI as an existing problem rather than a future issue. Participant 1 stated that the risk is "already confirmed, it's a fact" while Participant 5 presented it as "already happening". Perceived susceptibility was mainly discussed at a societal level. Participants saw biased AI as relevant in recruitment, policing, image generation, and decision systems. Participant 2 specifically highlighted recruitment as a context where AI screening could cause harm, warning that employers might use AI for pre-selection and "lose a lot of talent" because the system looks for something too narrow. This suggests higher personal susceptibility in employment-related contexts, even though most participants otherwise spoke about discrimination as affecting society or other groups rather than themselves directly. Within TTAT, framing the risk as already occurring clearly indicates a high susceptibility even if the

concern rarely was addressed to the participants personally but more often other groups.

The risk was rarely seen as very severe and often described with some ambivalence, for example Participant 1 described it as “maybe it’s less good?”. Several participants compared AI bias to human bias. Participant 3 argued that “everything is biased because everything is built by humans and humans are biased,” suggesting that AI does not introduce bias from nowhere but reproduces biases already present in society. Participant 5 also emphasised that humans cannot fully escape their own prejudices, saying: “We can’t get away from our prejudices, whether we use AI or not”. Using TTAT this can be analysed as a moderate to low perceived severity moderated by the relative framing where the risk from AI was not viewed as worse than the status quo.

Within a TTAT framing, perceived susceptibility was high because participants described discrimination as already occurring rather than as a hypothetical future risk. However, this susceptibility was mainly understood at a societal or group level, with participants more often referring to harms affecting others, institutions, or marginalised groups than themselves personally. Perceived severity was low to medium. Although participants recognised discrimination as problematic, its seriousness was moderated by a relative comparison with human decision-making. Since human bias was seen as already present in the alternative, AI bias was often interpreted less as a new or exceptional threat and more as a continuation or amplification of existing social bias.

5.1.2 Privacy and security

Participants acknowledged privacy and security as relevant AI risks, but their threat appraisals differed. Participants varied in their accounts of being exposed to the risk. Participant 7 described being ambivalent, saying there is “absolutely a risk,” but also hoped that their own data would “drown in all other information”. In contrast, Participant 2 emphasised individual vulnerability, noting that users may unknowingly share personal information in prompting, making it a more personal security

issue. Within TTAT, this can be seen as differing views of perceived susceptibility.

Perceived severity was also context-dependent. Participant 5 downplayed the additional AI-specific risk by arguing that "the average person does not have to worry about AI leaking their data because it's already leaked," while Participant 6 similarly compared AI data use to long-standing data collection by companies such as Google. However, severity increased in institutional contexts. Participant 5 identified the "biggest actual problem" as AI used by healthcare, banks, or systems containing large registers of people, while Participant 4 linked AI privacy risks to cybersecurity and the added complexity that can create mistakes or leaks. Thus, in TTAT terms, severity was moderate overall but higher where sensitive data or critical systems were involved, while also being moderated by comparison to existing privacy and security risks. This is similar to the pattern of relativising the risk found in the appraisal of risks from discrimination and bias as well.

There is also substantial divergence in mental models of how AI systems handle data between participants. Participant 3 asserted "the AI technology cannot leak anything because AI systems are bits of code that can't remember anything" and instead attributed all risk to the organisations operating them rather than the technology itself. In contrast, Participant 2 assumed persistent storage and potential reuse of all shared data, and prioritised privacy as her top concern. This variation indicates that mental models of AI shape both perceived severity and susceptibility.

Within a TTAT framing, perceived susceptibility ranged from low to high. It was lower when participants saw their data as already exposed or hidden within large datasets, but higher when they focused on users unknowingly sharing personal information or companies storing and exploiting data. Perceived severity was generally moderate, but increased in institutional contexts involving sensitive data, such as healthcare, banking, or cybersecurity. Overall, privacy and security risks were seen as real, but often relativised through comparison with existing digital systems.

5.1.3 Misinformation

Misinformation-related risks were generally perceived as both severe and likely, especially in relation to hallucinations. Participant 3 described incorrect AI outputs as inherent to the technology, arguing that AI "will always make these mistakes" and that this is "built into how AI technology works." Participant 7 similarly saw hallucinations as difficult to remove and considered them especially dangerous when combined with sycophantic design, since users may develop "a higher belief in the answers we get" and stop reflecting critically. Participant 2 also described hallucinations as common, stating that "everyone has probably encountered that AI makes things up."

Within TTAT, perceived susceptibility was therefore high, particularly for users who rely on AI without checking its outputs. Participant 6 emphasised this point, arguing that if people use AI answers without judging or analysing them, "that's the real risk." Participant 5 made a similar distinction between technically informed and uninformed users, saying it is difficult to understand how people can "blindly trust language models," but also noting that this happens because education about AI's limitations is lacking.

Perceived severity was also high because misinformation was not only seen as incorrect information, but as psychologically persuasive. Participant 7 explained that even when users know AI is "just code," the chat format can make them feel as if it reasons or thinks. This makes misinformation more convincing. Participant 4 similarly argued that people may trust AI "a little too much," comparing it to trusting a friend even when that friend may not actually know the answer.

At the societal level, Participant 7 highlighted epistemic erosion as a major risk, warning that people may begin to question whether news or political statements are real at all. Participant 2 also connected misinformation to AI-generated images in news, saying users "really have to watch out for what is real and not." Participant 5 added that AI-driven social media systems can reinforce filter bubbles by feeding users "what we want to see," making it harder to maintain a broad range of sources.

However, severity was somewhat moderated by comparison to existing misinformation and human fallibility. Participant 3 argued that humans can also misunderstand, lie, or give wrong answers, meaning AI is not entirely unique in producing false information. Participant 4 similarly framed AI as amplifying an existing problem, since there is already "so much misleading information" online. Overall, in TTAT terms, misinformation showed high perceived susceptibility and generally high severity, especially for less informed users and society at large, but this severity was partly relativised through comparison with human error and the existing online information environment.

5.1.4 Malicious actors and misuse

Malicious use was generally described as highly likely, and in some cases unavoidable. Participant 3 stated plainly that "it will happen," arguing that little can be done because AI models are openly available and safety restrictions can be removed. Participant 1 similarly saw AI as a tool that will be used for both good and bad purposes, but expected negative use to dominate, at least initially, because of human "greed." Participant 7 also described misuse as already partly occurring, especially in relation to manipulation and possibly cyberattacks. Within TTAT, this indicates high perceived susceptibility. The risk was not seen as distant or speculative, but as already emerging or inevitable. Susceptibility was mainly located at a societal level, as participants discussed cyberattacks, fraud, terrorism, deepfakes, and manipulation as risks affecting many people rather than themselves personally.

Perceived severity was moderate to high, but often framed through amplification rather than novelty. Participant 2 argued that AI makes cyberattacks and fraud more accessible because "with such a smart assistant you can achieve more," especially at a larger scale. Participant 1 similarly argued that AI has already amplified cyberattacks. Deepfakes and synthetic media were also seen as serious risks: Participant 1 warned that AI can create "material that is completely false," while Participant 5 argued that fake image, video, and voice content can make it difficult to know "what is real, what is not real." Within TTAT, perceived severity was therefore moderate to high. It was moderated by the fact that participants

often saw these harms as extensions of existing problems, such as fraud and cyberattacks, rather than entirely new threats. However, severity increased because AI was seen as making these harms easier, faster, more scalable, and accessible to less skilled actors.

Within TTAT, malicious misuse involved high perceived susceptibility and moderate-to-high perceived severity. Severity was partly relativised because participants saw these harms as extensions of existing crime and manipulation, but it remained high where AI was perceived to increase scale, accessibility, and effectiveness.

5.1.5 Human–computer interaction

Human–computer interaction (HCI) risks were highly salient and grounded in everyday experiences, especially education, social media, and reliance on AI tools. Perceived susceptibility was generally high because AI was described as already embedded in daily routines. Participant 5 connected this to social media, arguing that AI-driven apps are designed to keep users engaged and can become "a form of addiction." Participant 7 similarly described a growing dependency on AI systems, suggesting that they may be cheap or free now because users are being encouraged to become reliant on them.

Education was the most central context. Participant 5 described AI in education as one of the biggest risks, arguing that schools, teachers, and students do not yet know how to handle it. They especially worried about children, asking "how will children know how dangerous it can be?" Participant 1 also framed education as the area where AI could become a "death blow," based on seeing coursemates use AI "not to understand or become better, but just to get by." Participant 3 made a similar point in programming education, arguing that students who let AI write code "do not learn" the skills they need.

Within TTAT, this indicates high perceived susceptibility, but the concern was mainly directed toward other groups rather than the participants themselves. Younger users, schoolchildren, less technically knowledgeable users, and other students were often framed as most exposed. Participant 5

suggested that university students in their own programme may be more reflective, while younger children are more vulnerable. Participant 2 and Participant 7 also emphasised that people with less knowledge of AI may be more susceptible to overtrust and misuse.

Perceived severity was also high, but mainly because of gradual long-term effects rather than immediate harm. Participant 7 warned that AI may affect "our ability to think and our intelligence," while Participant 6 argued that if AI is used to "fill in the blanks," people may lose analytical reasoning and become unable to think independently if the tool disappears. Participant 3 extended this to the labour market, warning that overreliance may produce workers who believe they know more than they actually do because they have outsourced the technical work to AI.

Overreliance and misplaced trust were central. Participant 7 argued that people can have "very high trust in AI," partly because it appears more capable than it is. Participant 4 similarly explained that AI systems can feel trustworthy because they answer quickly and confidently, making outputs seem more reliable than they are. Participant 7 also noted that people already use AI as an excuse, saying "no, it was AI," which suggests responsibility diffusion.

Within TTAT, HCI-related risks involved high perceived susceptibility and high perceived severity. However, susceptibility was usually projected onto others rather than framed as a direct personal vulnerability. The threat was not mainly sudden harm, but gradual deskilling, reduced critical thinking and weaker educational outcomes.

5.1.6 Socioeconomic and environmental harms

Socioeconomic risks, particularly those related to employment, emerged as highly salient in participants' threat appraisals. The impact of AI on the labour market for software developers was brought up spontaneously by all participants without prompting, indicating that this is a central and immediately relevant concern. However, both perceived severity and perceived susceptibility varied considerably, and were often characterised

by uncertainty and competing interpretations rather than stable evaluations.

Socioeconomic risks were mainly discussed through employment, especially entry-level software development. Participant 2 described how junior system developer roles may "disappear a little," since AI is already good at debugging and small fixes. Participant 5 similarly argued that junior programming jobs may soon decrease because AI can do simpler programming tasks "much faster" than junior developers. Participant 3 expressed stronger personal concern, saying they would have felt safer entering the industry "two years earlier," before junior jobs began to be cut. Participant 7 also worried that easier entry-level roles may be replaced, creating a gap between graduates and more experienced workers.

Within TTAT, perceived susceptibility was therefore moderate to high, especially for those entering the labour market. However, susceptibility was often differentiated between self and others. Participants enrolled in AI-focused programmes, or those who positioned themselves as capable of working with AI systems, such as Participant 2, tended to report lower personal susceptibility. These participants saw themselves as partly protected by AI-related competence, while expressing stronger concern for future students, junior developers, marketing students, or less specialised workers.

Perceived severity was highest when participants focused on long-term labour market consequences. Participant 3 argued that if companies remove junior positions, the question becomes: "What happens in ten years?" when senior workers leave and there are fewer trained replacements. Participant 2 also suggested that where four programmers were previously needed, perhaps only one person will be needed to supervise AI. However, severity was moderated by the belief that AI will not replace all work. Participant 1 rejected the idea that software development will "die out," arguing that human knowledge is still needed for full-scale projects. Participant 5 similarly believed that jobs requiring human contact, such as healthcare, will remain because people still trust and need human interaction.

Several participants also located socioeconomic risk in wider systems rather than AI alone. Participant 4 argued that job displacement is driven less by AI itself than by the "capitalist system" that replaces humans with AI for efficiency. Participant 1 saw AI as potentially contributing to centralised power, while Participant 4 raised concerns about unequal distribution of benefits, where profits may end up with "some CEO" rather than those doing the work.

Within TTAT, perceived severity was moderate to high for employment risks, especially regarding junior roles and long-term skill pipelines. Perceived susceptibility was high for certain groups, such as less experienced junior software developers. This was also reflected in higher personal susceptibility among participants with less AI training, while those with more AI education saw themselves as less personally susceptible.

5.1.7 Safety and limitations

Risks related to advanced AI, such as misalignment, loss of control, autonomous decision-making, or machine consciousness, were generally not perceived as immediate. Participant 1 believed such developments might happen eventually, but not soon, arguing that current technology and resources are still limiting factors and that it might take "50, 100 years and forward." Participant 7 similarly saw "some possibility" of advanced AI emerging, but "not with the systems we have now." Participant 6 also rejected this as a near-term risk, arguing that current AI architectures do not support consciousness and that a different kind of technology would be needed.

Within TTAT, this indicates low perceived susceptibility. Participants did not see themselves or society as currently exposed to advanced AI safety risks. When acknowledged, the risk was usually placed in the distant future and made dependent on uncertain technological breakthroughs, which reduced urgency compared to more immediate risks such as misinformation, overreliance, or labour market disruption.

Perceived severity was more ambiguous. Participant 5 argued that a powerful AI with influence over the real world would be "definitely more worrying," especially if humans could not understand why it made certain decisions. Participant 4 also saw potential danger in autonomous military systems, arguing that humans should "never ever have an AI system that makes the decision instead of us." Participant 2 described AI takeover scenarios as unlikely but not impossible, noting that development feels difficult for humans to keep up with. Within TTAT, perceived severity could be high in hypothetical scenarios involving autonomy, infrastructure, or military decision-making.

Participant 3, who primarily referred to large language models, argued that consciousness in AI is impossible because an AI system only functions by predicting the next word. In this account, the conceptual focus on specific types of AI systems appears to constrain which risks are considered plausible or relevant, thereby narrowing the scope of perceived threats in a similar way as discussed in section 5.1.2 on privacy risks.

Overall, within a TTAT framing, safety and limitation risks were characterised by low perceived susceptibility but potentially high perceived severity, meaning the overall threat appraisal remained weaker than for more immediate AI risks.

5.2 Risk prioritisation

Across participants, the most consistently prioritised category of risk was human-computer interaction. It was the top concern for all but Participant 2. In particular, different forms of overreliance on AI systems emerged as a central concern. This included worries about reduced individual capabilities, uncritical trust in outputs, and challenges in educational contexts where AI may substitute rather than support learning.

In addition to these risks, a few more were presented as highly prioritised. However, there was substantial variation in which specific risks were prioritised within and across participants. The following overview presents the participants' prioritisation in addition to HCI risks:

- Privacy related risks, particularly concerning personal data, were the top concern of Participant 2.
- Socioeconomic risks, including distorted public discourse and impacts on the labour market, were ranked highly for Participant 3.
- Risks related to AI in education and addictive patterns of use, especially in relation to social media, were emphasised by Participant 5.
- Participant 7 prioritised risks related to misinformation, specifically the risk that AI generated content may result in distrust in other sources of information.

These additional risks varied substantially between participants, while concern around HCI was more consistent, particularly in relation to how AI systems shape human reasoning, behaviour, and learning.

The justification for what made different risks important varied across participants. Two participants explicitly grounded their prioritisation in assumptions about human self-interest. Participant 1 argued that people in general tend to prioritise risks that affect them personally, particularly in relation to protecting their own data and security. Participant 3 similarly emphasised that his primary evaluation criterion was how risks would affect him directly, especially in terms of his skills and future prospects. At the same time, he reflected that a more appropriate method would be to adopt a broader consequentialist perspective, assessing risks based on how many people are affected and the severity of the impact. Other participants appeared to base their prioritisation on what they perceived as neglected or insufficiently addressed areas, rather than on direct personal impact.

5.3 Problem-focused coping appraisal and responses

This section covers the participants' reasoning about problem-focused coping responses. Some of the suggested responses were discussed in relation to particular risks while others were considered more broadly. Most can, however, be understood in relation to the participants' high-priority risks, as this is where their reasoning tended to focus. Table 4

summarises the dominant patterns in participants' coping appraisal of problem-focused strategies. The table should be read as an analytical overview rather than a quantitative ranking, since participants differed in how they evaluated the effectiveness, costs, and feasibility of each strategy.

Problem-focused coping strategy	Main actor(s)	Perceived effectiveness	Perceived costs	Self-efficacy	Key modifiers of appraisal
Educate about AI	Individuals, peers, schools, authorities, media	High, but incomplete	Low–high depending on depth	High individually/interpersonally; lower societally	Basic awareness vs deep expertise; psychological tendencies; unequal access
Responsible AI use	Individual users, students, professionals, organisations	High when users have domain competence and verify outputs	Moderate; time, effort, reduced efficiency	Moderate–high for personal use; lower under social/work pressure	Convenience pressure; domain competence; high-stakes contexts
Regulating AI	Governments, EU, international bodies, authorities	High in principle, uncertain in practice	Moderate–high; slower innovation, enforcement burden	Low individually; higher for institutions	Global coordination, loopholes, decentralised development
Designing safer AI systems	Developers, companies, researchers	Moderate; useful for bias, transparency, safety, but limited against misuse	Moderate–high; technical complexity, data trade-offs, performance limits	Low for ordinary users; higher for technical specialists	Open models, adversarial users, data access, explainability
Developers managing AI risk	Developers, AI companies, project leads	Low–moderate; possible but not trusted as sufficient	Moderate; conflicts with incentives and role boundaries	Low–moderate; limited individual developer power	Profit incentives; unclear normative responsibility; neutral tool view
Mitigating risk in career and research	Students, future developers, researchers	Moderate; potentially meaningful in specific domains	Moderate–high; career constraints, limited broader influence	High for some with concrete projects; uncertain broadly	Career agency; specialisation; mismatch with wider AI acceleration
Addressing systemic drivers of AI risk	Governments, society, institutions, economic actors	High in theory, abstract in practice	High; requires large-scale political/economic change	Low individually	Economic incentives, automation pressure, collective action problems

Table 4. Summary of participants' problem-focused coping appraisals.

5.3.1 Educate about AI

The most prominent problem-focused coping strategy was increasing knowledge about AI. Participant 4 argued that many people treat AI as a "magic box" because they do not understand how it works, leading them to trust it too much. They therefore emphasised teaching people that AI is "not a soul in the computer," but a system producing statistically likely answers. Participant 6 similarly argued that "awareness is really important" and that children should learn how LLMs work before deciding "what they want to trust." Participant 5 called for courses in AI use, including how to use AI "in a moral way" and how teachers should adapt education to AI.

Within TTAT, education functions as problem-focused coping because it aims to reduce AI-related risks by changing how people use, trust, and interpret AI systems. Rather than avoiding AI, participants framed knowledge as a way to make use safer, more critical, and more deliberate.

Perceived effectiveness was generally high, but limited. Participant 4 compared AI education to learning safety rules for a bandsaw: understanding the tool helps users avoid harm. Participant 7 also believed knowledge makes it easier "not to be manipulated and fooled," especially by hallucinations or sycophantic responses. However, Participant 7 also noted that even when one knows AI is "just code," it can still feel like "something more," suggesting that psychological tendencies may limit education's effectiveness. Within TTAT, this indicates relatively high perceived effectiveness, but not complete protection. Education was seen as reducing susceptibility, while not fully eliminating overtrust or anthropomorphising.

Perceived costs varied. Participant 4 saw sufficient understanding as challenging but surmountable. Participant 2 noted that AI changes "so damn fast" that it is difficult to keep up, especially for people who are not already interested. Participant 7 added that learning takes time and that not all resources are freely accessible. Participant 3 described deeper understanding as requiring "tens of thousands of kronor plus hundreds or thousands of hours," making it unrealistic for most individuals. Within TTAT, perceived costs were therefore low to moderate for basic awareness,

but high for deeper technical understanding. This makes education unevenly accessible.

Self-efficacy was strongest at the individual and interpersonal level. Participant 1 described taking extra AI-related courses because “knowledge is power.” Participant 5 positioned themselves as a bridge between the technological and human sides of AI, while Participant 7 described short conversations with friends as helpful. However, Participant 3 argued that ordinary individuals cannot be expected to handle this alone and that larger organisations with expertise are needed.

Overall, education was the clearest form of problem-focused coping in the data. It was perceived as relatively effective because it could reduce overtrust, misinformation, and unsafe use, but its effectiveness was limited by psychological biases and the complexity of AI systems. Its costs ranged from low for basic awareness to very high for deeper expertise. Self-efficacy was strongest for personal learning and peer discussions, while large-scale public education was seen as requiring schools, authorities, media, or expert organisations.

5.3.2 Responsible AI use

Responsible use was another central problem-focused coping strategy. Participants described it as using AI selectively, critically, and with continued human judgement. Participant 6 argued that AI should not be used “without a clear objective,” but only as a tool for specific tasks where the user already understands the broader project. Participant 1 made a similar distinction in education, saying students should “see it as a tool” rather than “a way to get by.” Participant 3 also argued that AI should not replace the work needed to build competence, especially in programming education, because doing the work oneself “unlocks that possibility in the head” for future use.

Within TTAT, responsible use functions as problem-focused coping because it aims to reduce exposure to AI-related risks through controlled interaction. Rather than rejecting AI entirely, participants described ways of limiting when and how it is used, especially to avoid overreliance, misinformation, skill loss, and weak accountability.

Perceived effectiveness was relatively high when responsible use involved domain competence, verification, and human oversight. Participant 6 argued that if users cannot "judge or analyse the results," then AI should not be used in that way. Participant 2 made a similar point in relation to programming assignments, noting that even if AI writes code, students must still defend and explain it, and "it is your responsibility." Output verification was also seen as important. Participant 4 suggested asking multiple AI systems or comparing answers, while Participant 3 described restarting the chat or asking another AI system to see whether the same answer appears again. Human-in-the-loop control was especially emphasised. Participant 3 argued that there must "always be a human" checking AI decisions because "you cannot hold an AI responsible," while Participant 4 stated that humans should "never ever have an AI system that makes the decision instead of us."

Within TTAT, perceived effectiveness was linked to the belief that responsible use increases avoidability by preserving human judgement and accountability. The strategy was seen as most effective when users had enough knowledge to evaluate outputs and when AI remained advisory rather than autonomous. However, within TTAT, perceived effectiveness was lower in situations where users lacked domain knowledge or where AI was used in high-stakes areas such as healthcare, recruitment, or education.

Perceived costs were mainly linked to time, effort, and reduced efficiency. Participant 7 noted that trying to do as much as possible independently takes longer and can make one appear less capable because others use AI to work faster. They also described how limited use can gradually expand: one may begin by asking AI to fix sentences, but then "suddenly" ask it to write a whole section.

Within TTAT, perceived costs were therefore not negligible. Responsible use required time, restraint, critical checking, and sometimes accepting slower performance. The convenience of AI created pressure against responsible use, making the strategy harder to sustain even when participants viewed it as desirable.

Self-efficacy varied, but was generally strongest for personal use. Participant 1 described deliberately limiting AI in studies because they wanted to "become as good as possible." Participant 6 described reading documentation, checking accuracy, and deciding whether a tool is a good choice before using it. Participant 5 also emphasised thoughtful use at the organisational level, arguing that companies should hire people who understand both AI and its limitations rather than applying AI "everywhere just because it is called AI." Participant 7, however, suggested that responsible use is difficult because people are "a little forced to integrate AI" to keep up with the pace.

Within TTAT, self-efficacy was moderate and strongest for personal AI use, especially through selective use, source criticism, and verification. However, self-efficacy was weakened by social and institutional pressures, such as speed expectations, workplace adoption, and the difficulty of resisting convenient tools.

Overall, within TTAT, responsible use was framed as a practical form of problem-focused coping. It increased perceived avoidability by keeping humans in control, restricting AI use to suitable domains, verifying outputs, and maintaining accountability. Its main limitation was not that participants doubted the strategy in principle, but that it required sustained effort, competence, and resistance to convenience and social pressure.

5.3.3 Regulating AI

Regulation was another problem-focused coping strategy, but participants described it as both necessary and difficult. Participant 2 argued that "real laws" are needed globally, not only in Europe, because regional rules can be bypassed. Participant 3 similarly suggested that smaller governments lack the resources to manage AI risks and that regulation should ideally come from an international level, such as the UN. Participant 4 emphasised the need to pause and reflect, arguing that society must sometimes "stop, look left and right" rather than letting technology push everyone forward too quickly. Within TTAT, regulation functions as problem-focused coping because it targets AI risks structurally rather than individually. Slowing

development, creating legal frameworks, and increasing oversight were seen as ways to increase avoidability before harmful systems are deployed.

Perceived effectiveness was mixed. Participant 5 argued that development needs to "slow down a little" so society has time to understand the consequences. Participant 2 similarly argued that faster development creates more mistakes and that it may be better to "slow down a little and actually think." Participant 4 believed that with "slow steps, safeguards, reconsideration and critical thinking," errors can be stopped. However, Participant 2 also noted that restrictions are difficult because "there are always loopholes," while Participant 6 argued that "no one can manage AI" because independent researchers can build systems globally. Within TTAT, perceived effectiveness was therefore high in principle but limited in practice by enforcement problems, loopholes, and decentralised development.

Perceived costs were mainly linked to slower innovation and lost opportunities. Participant 2 acknowledged that regulation may prevent AI from developing "even faster" or "even better," and that society pays a cost by not simply pushing ahead. Within TTAT, perceived costs were moderate to high, since regulation was seen as protective but also potentially delaying benefits.

Self-efficacy was generally low. Participant 2 found it hard to imagine world leaders agreeing on global AI regulation, while Participant 6 stated that AI is too decentralised to control. Participant 3 placed responsibility mainly on large actors such as the EU, US, China, or international institutions, rather than individuals. Within TTAT, self-efficacy was low at the individual level and higher only when imagined through states, authorities, or international organisations.

Overall, within TTAT, regulation was framed as a problem-focused coping strategy with high theoretical value but limited practical confidence. It could increase avoidability by slowing development, enforcing safeguards, and coordinating standards, but its effectiveness was weakened by globalisation, loopholes, decentralised research, and weak individual self-efficacy.

5.3.4 Designing safer AI systems

Designing safer AI systems was described as a problem-focused coping strategy aimed at reducing risks through the technical structure of AI itself. Participant 4 discussed technical safeguards, noting that companies such as OpenAI and Gemini have built "walls" to prevent harmful outputs, but also emphasised that users can still find loopholes. Participant 3 was more sceptical, arguing that if models are openly available, people can "just remove the safety parts afterwards." Participant 2 connected safer systems to training, but noted that AI "can technically always become biased". Within TTAT, designing safer systems functions as problem-focused coping because it attempts to reduce the threat at its source, by making AI less biased, less harmful, more transparent, or harder to misuse.

Perceived effectiveness was mixed. Participant 4 described their own work on mitigating the "black box problem" in computer vision by making it easier to trace why an AI system made a certain judgement. Participant 6 similarly suggested that bias could be reduced through "other AI research" using different data or "a different architecture to make a decision." However, both Participant 4 and Participant 3 stressed that safeguards are weaker against deliberate misuse, since loopholes can be found and open models can be modified. Within TTAT, perceived effectiveness was therefore moderate. Safer design was seen as useful for transparency, accountability, and reducing some bias or error, but limited by adversarial users, open models, and the difficulty of preventing all misuse technically.

Perceived costs were mainly linked to trade-offs between safety, performance, and data access. Participant 6 described data restrictions as a dilemma: more data can improve accuracy, but using restricted or private data creates privacy concerns. If data is limited, some accuracy may be lost. Within TTAT, perceived costs were moderate to high because safer systems may require architectural redesign, controlled data access, more development time, and possible compromises in accuracy or capability.

Self-efficacy was generally low because safer system design was located with developers, companies, and researchers rather than ordinary users. Participant 4 was an exception, describing their own research on

explainable computer vision systems. Participant 6 also framed safer AI as a research problem, while Participant 2 expressed limited confidence in training-based solutions.

Overall, within TTAT, designing safer AI systems was seen as potentially valuable problem-focused coping, but with uncertain practical success. It could increase avoidability through transparency, accountability, data governance, and better model architecture, but was limited by technical complexity, data trade-offs, loopholes, open models, and low individual influence.

5.3.5 Developers managing AI risk

Participants generally did not view AI developers as a sufficient locus of responsibility for managing AI risks. All participants discussed possible actions that developers could take to mitigate AI risks, but these actions were often viewed as unfeasible or insufficient. The reasoning behind this differed somewhat. Some participants approached the issue from a practical perspective based on incentives: developers and companies were seen as having incentives that are not aligned with risk management, and were therefore not expected to prioritise mitigation. Others approached the issue from a more normative perspective, arguing that it is not the role of developers to determine what is right or wrong.

5.3.6 Mitigating risk in career and research

A smaller subset of participants described how they could contribute to mitigating AI-related risks through their future careers and academic work. These strategies reflect an orientation toward influencing AI development and deployment at a broader level.

Participant 4 described actively working on a technical solution to the "black box" problem in AI through his thesis project. By developing architectural approaches that make AI decision-making more transparent and traceable, he aimed to address issues related to accountability, safety, and legal responsibility, particularly in high-stakes contexts such as autonomous systems.

Participant 5 expressed a desire to shape her future career in a way that does not contribute to accelerating AI development, but instead focuses on responsible application. She considered both academic and industry paths where her technical competence could be used to critically assess AI systems, highlight their limitations, and ensure that they are not indiscriminately applied. She also emphasised that many peers in her AI-focused programme appeared more interested in accelerating AI development than in critically evaluating its societal consequences. As a result, she did not view computing students as a group as likely to collectively mitigate AI-related risks in the future.

Participant 7 described a thesis project focused on understanding how professionals use AI in practice and how such use affects them, both positively and negatively. The project also aimed to develop more structured methods and best practices for AI use. Although the participant did not explicitly frame this work as AI risk mitigation, the project focused on reducing unstructured or overly reliant forms of AI use through more reflective practices.

Within TTAT, these actions can be understood as forms of problem-focused coping, where participants seek to reduce perceived risks through direct intervention, structured practices, or responsible development. Participants 4 and 7 had already begun engaging in such behaviours, suggesting relatively high coping appraisals and perceived self-efficacy regarding their ability to contribute to mitigation efforts. Participant 5, in contrast, expressed greater uncertainty regarding the broader impact such efforts could have within the wider AI development environment.

5.3.7 Addressing systemic drivers of AI risk

A smaller but distinct coping strategy involved addressing the broader systems that shape how AI is developed and used. Participants 1 and 4 located some AI-related risks in societal and economic structures rather than in AI itself. Participant 4 argued that AI is “only the tool” and that the larger problem lies in the economic system that rewards efficiency, automation, and replacing human labour. In this view, coping requires changing the conditions that make harmful AI use attractive. Within TTAT,

this functions as problem-focused coping because it targets the underlying drivers of AI risk rather than only managing individual use or technical design. The aim is to increase perceived avoidability at the structural level by changing incentives around deployment, automation, and profit.

Perceived effectiveness was high in principle but abstract. Participant 4 suggested that risks such as job displacement and unequal benefit distribution could be reduced by rethinking economic incentives, including redistributive mechanisms or limits on income accumulation. Participant 1 also linked AI to broader centralisation of power, suggesting that AI may support an already existing direction rather than create it alone. Within TTAT, perceived effectiveness is potentially high because systemic change addresses root causes. However, this effectiveness remained speculative because participants did not present clear or immediately actionable mechanisms.

The perceived costs were high. These strategies required large-scale political, economic, or collective change rather than small individual adjustments. Participant 4 framed the issue as no longer only a conflict with one employer or workplace, but with "the whole economy," which makes the coping strategy demanding and difficult to realise. Within TTAT, perceived costs are therefore high because systemic coping requires coordination, time, political will, and major institutional change.

Self-efficacy was low. Participants generally did not present themselves as able to directly change these systems. Participant 4 argued that individuals cannot change this alone, but that cooperation may create differences over time.

Overall, within TTAT, addressing systemic drivers was a problem-focused coping strategy with high theoretical effectiveness but low practical self-efficacy. It targets root causes of AI risk, but its costs and scale make it less immediately actionable than education or responsible use.

5.3.8 Additional personal strategies

In addition to the previously described forms of problem-focused coping, participants also engaged in a set of more individualised and strategic adaptations aimed at reducing their personal exposure to AI-related risks. One common approach involved developing skills or competencies in areas where AI was perceived to be less capable. For example, Participant 3 described focusing on highly specialised aspects of programming, such as performance optimisation and niche technical domains, where current AI systems are less effective. This was framed as a way to maintain relevance in a changing labour market and reduce the risk of being replaced.

Beyond career-related adaptations, Participants 2 and 5 described strategies for managing informational risks, such as actively diversifying their sources of information to counteract algorithmic filtering and bias. This included making a conscious effort to seek out different perspectives rather than relying solely on AI-curated or platform-driven content. Participant 3 emphasised the importance of selecting more trustworthy or controlled AI services, for example by using smaller providers or systems where data handling practices are perceived as more transparent.

Within TTAT, these strategies reflect relatively high self-efficacy and perceived effectiveness, as participants identify concrete actions they can take themselves to mitigate risk. At the same time, these strategies are primarily limited to the individual level and do not address broader systemic risks, instead focusing on reducing personal vulnerability.

5.4 Emotion-focused coping

In addition to problem-focused strategies, participants also engaged in emotion-focused coping to manage the uncertainty and anxiety associated with AI-related risks. These strategies did not primarily aim to reduce the risks themselves, but rather to regulate emotional responses and maintain a sense of control.

5.4.1 Cognitive reappraisal

Cognitive reappraisal appeared when participants reinterpreted AI-related risks in ways that made them more emotionally manageable. Participant 3 described avoiding general AI-related media and instead shifting the question inward: "How does this affect me?" and "What do I lose when I use AI? What do I gain?" This reframed AI risk from an overwhelming societal issue into a personal reflection on learning, skill development, and control. Participant 4 used a value-based reappraisal. While acknowledging fears around AI replacing programmers and designers, they emphasised that human creativity and effort still have unique value. They argued that creative work gains meaning because a human has "put in sweat and tears", something AI does not provide. Participant 7 similarly reduced privacy anxiety by hoping their data would "drown" in the mass of other information.

Within TTAT, cognitive reappraisal functions as emotion-focused coping because it changes the interpretation of the threat rather than reducing the risk itself. The risk is not denied, but its emotional salience is lowered by reframing it as personally assessable, socially familiar, or limited by the continued importance of human creativity and judgement.

5.4.2 Deliberate optimism and positive reframing

Deliberate optimism appeared when participants consciously focused on AI's positive possibilities to manage concern or uncertainty. Participant 5 described taking a positive stance when talking to people who are strongly against AI, partly to help them avoid "anxiety over AI taking over the world and killing everyone." Participant 4 similarly stated that they focus on "the positive sides" of AI, especially human-centred and specialised applications, rather than becoming absorbed by broader fears based on generalised advanced AI systems. Participant 6 also expressed this orientation, saying that "AI is there to help us, I only focus on the positive sides" and that they do not see many issues with it.

Participant 2 used positive reframing in relation to employment risks. Although they acknowledged that junior developer and marketing roles may be affected, they also hoped AI would create new jobs and

opportunities in technical fields, saying it would be good if the effects went "both ways."

Within TTAT, deliberate optimism functions as emotion-focused coping because it reduces the emotional salience of AI-related threats by shifting attention toward benefits, opportunities, and constructive uses. Unlike problem-focused coping, it does not directly reduce the risk itself. Instead, it helps participants maintain a more hopeful and manageable outlook in the face of uncertainty.

5.4.3 Fatalistic reasoning

Fatalistic reasoning appeared when participants framed certain AI-related risks as inevitable and beyond meaningful control. Participant 3 expressed this most clearly in relation to misuse, stating that "it will happen" and that there is "nothing we can really do to stop it." They linked this to the open availability of AI models, arguing that once model code is publicly accessible, safety restrictions can be removed and harmful use becomes unavoidable.

This reasoning did not deny the existence of the risk. Instead, it accepted the risk as a structural condition of AI development. Participant 4 expressed a similar view in relation to malicious use, noting that there would always be people who try to exploit systems and that there is no way to avoid this completely.

Within TTAT, fatalistic reasoning functions as emotion-focused coping because it manages the emotional burden of low perceived control. When participants see coping efficacy as low, accepting the risk as unavoidable reduces the pressure to identify active solutions. In this sense, fatalism does not necessarily indicate indifference. Rather, it reflects a way of stabilising emotional responses when AI risks are perceived as real but not realistically preventable.

5.4.4 Reframing AI as a neutral tool

A recurring pattern in the material concerns participants describing AI as a neutral tool whose effects depend on human use. This type of statement

can serve different analytical roles. In some cases, it appears as a general conceptual stance about the nature of technology. In other instances, however, it occurs in direct response to previously expressed concerns, suggesting a different function. For example, Participant 1 first acknowledged that the negative aspects of AI "remain in the back of [their] mind," but then immediately emphasised that "the technology itself is completely neutral." In this sequence, the neutrality framing can be interpreted as a form of emotion-focused coping, where the perceived threat is reduced by shifting attention away from the technology and towards human use. This illustrates how the same framing of AI may both reflect a genuine belief and function as a way to manage perceived risk.

5.4.5 Avoidance

Some participants engaged in avoidance strategies, particularly by limiting their exposure to AI-related media. While some justified this in terms of distrust toward media portrayals, in other cases it appeared to function primarily as a way to regulate emotional responses. For example, Participant 3 described avoiding AI-related media because it made them feel anxious about their future career. In such cases, avoidance reduces exposure to threat-related information rather than addressing the underlying risk. Within TTAT, avoidance functions as emotion-focused coping because it reduces exposure to threat cues rather than addressing the underlying risk. By avoiding AI-related media, participants can lower anxiety and regain emotional stability, but the strategy does not increase actual avoidability or reduce the AI-related threat itself.

5.5 Cross-cutting influences on risk perception

This section presents cross-cutting factors that shaped participants' appraisals across multiple risk categories and coping strategies. Rather than being tied to one specific AI risk, these themes influenced how participants understood the nature of AI-related threats, who they believed was exposed or responsible, and which responses appeared realistic. The section therefore connects the category-specific findings above by showing how responsibility attribution, relative comparisons, scope of concern,

mental models of AI, and social influence shaped both risk appraisal and coping appraisal.

5.5.1 Responsibility

Responsibility attribution emerged as a central cross-cutting factor shaping how participants evaluated both risks and possible responses. A recurring pattern was a distinction between who is currently in a position to act and who ought to be responsible for managing AI-related risks. This distinction structured coping appraisal by influencing which problem-focused strategies participants considered realistic.

AI developers were rarely treated as a sufficient locus of responsibility. Participant 2 argued that developers "need to take responsibility," but also that "it is not really their place to decide what is okay." Participant 5 made a similar distinction, arguing that individual developers mainly follow orders and that responsibility lies more with company leadership or authorities. Companies were also treated with scepticism. Participant 3 stated that "we cannot rely on companies to do this risk analysis," while Participants 4 and 5 argued that economically driven companies are unlikely to implement safeguards unless required to do so. As a result, coping strategies that rely on voluntary developer or company action were often viewed as weak or unrealistic.

This reluctance to locate responsibility with developers and companies also appeared connected to a broader framing of AI as a neutral tool. Participants 1, 3, 4, 5, and 6 all, in different ways, described AI as something whose effects depend primarily on how humans use it. Participant 1 stated that "the technology itself is completely neutral," while Participant 4 argued that "AI is only the tool" and that harms such as job displacement are more directly caused by the economic system in which AI is deployed. Participant 5 similarly compared AI to earlier tools such as calculators, suggesting that the technology itself does not necessarily make humans worse, but changes what humans can focus on. In this framing, responsibility shifts away from the technical artefact and often away from the people building it, toward users, organisations, economic incentives, or authorities that decide how AI is applied. This helps explain why

developers were often seen as influential but not primarily responsible: if AI is understood as a neutral tool, then the main moral responsibility is placed on those who deploy, govern, or misuse the tool rather than on those who create it.

Responsibility was instead commonly attributed either to end users or to governments and regulatory authorities. Participant 7 argued that individuals still need to take responsibility and "not just hope someone else will solve it." However, individual responsibility was often framed as insufficient. Participant 5 noted that responsibility currently lies too much with private users, even though many people lack enough knowledge to understand what AI systems they are exposed to. Participant 3 similarly argued that it is unrealistic to expect ordinary individuals to reason well about AI, since understanding it requires substantial time, money, and expertise.

Consequently, several participants positioned governments or supranational authorities as the actors that should carry primary responsibility. Participant 2 argued that AI requires "real laws," not only in Europe but globally, because regional regulation can be bypassed. Participant 3 preferred larger institutions such as the EU or UN, since smaller governments lack the resources to manage AI risks effectively. Participant 7 also saw regulation and societal-level support as necessary, while emphasising that AI's global and decentralised character makes national regulation difficult.

This configuration of responsibility attribution has clear implications for coping appraisal. When responsibility was assigned to actors perceived as unwilling or unable to act, such as profit-driven companies or practically constrained governments, confidence in problem-focused coping weakened. When groups such as developers were not seen as responsible, participants were less likely to assign problem-focused coping to them. Participants therefore moved between calls for regulation, personal responsibility, and more limited individual strategies such as education, responsible use, and source criticism. Thus, responsibility attribution did

not only reflect views about governance; it also shaped perceived coping efficacy.

5.5.2 Relative nature of risk appraisal

A recurring pattern was that participants evaluated AI-related risks in relation to existing systems rather than in isolation. Instead of asking only whether AI introduces harm, they often assessed whether AI makes existing harms worse than the current alternative. This relative framing appeared across discrimination, privacy, misinformation, and misuse.

For discrimination, Participant 3 argued that "everything is biased because everything is built by people and people are biased," while Participant 5 similarly stated that humans cannot escape prejudice "whether we use AI or not." In this framing, AI bias was not seen as a completely new problem, but as a continuation of bias already present in human data and decision-making. This did not make the risk irrelevant, but it reduced its perceived severity.

A similar pattern appeared in privacy. Participant 5 argued that the average person does not need to worry too much about AI leaking their data because it is "already leaked" through existing digital systems. Participant 6 also compared AI-related data risks to earlier forms of data collection, such as search engines collecting user data. Here, AI was interpreted as part of an already compromised data environment rather than as a fundamentally new privacy threat.

This relative appraisal also appeared in relation to misinformation and misuse. Participant 3 compared AI hallucinations to human fallibility, arguing that humans can also misunderstand, lie, or give wrong answers. In misuse, Participant 2 framed AI-enabled fraud and cyberattacks mainly as scaling up existing harms, rather than creating entirely new ones. The concern was therefore not always the existence of fraud or manipulation itself, but that AI could make it easier, faster, or more widespread.

Within TTAT, this comparative reasoning mainly shaped perceived severity. When AI-related harms were seen as continuous with existing problems, their severity was often moderated, even when participants still

acknowledged them as real. However, relative appraisal could also increase severity when AI was seen as amplifying scale, speed, or accessibility. Thus, participants' threat appraisals depended not only on what AI can do, but on whether it was perceived as worse than, similar to, or merely faster than existing systems.

5.5.3 Variation in scope of concern

Participants shifted between different scopes of concern when evaluating AI risks. They did not assess risk from one stable position, but moved between the self, close social circles, vulnerable groups, and society more broadly. Participant 3 was most explicit about self-oriented prioritisation, stating that the main question was "how will this affect me?" especially regarding their own skills and future career. Participant 2 also discussed risks partly through personal exposure, particularly privacy and job-market concerns.

However, concern often extended beyond the self. Participant 5 emphasised that AI affects different groups differently, including younger generations, pupils, university students, and older people. Their strongest concerns were often directed toward children and education rather than themselves. Participant 7 similarly focused on less knowledgeable users, future graduates, and people who may overtrust AI systems.

Temporal scope also mattered. Immediate and visible risks received the most attention, especially educational overreliance and junior labour-market disruption. Participant 5 described AI in education as something that is "already happening," while Participant 7 emphasised present risks around overtrust and dependency. Long-term risks such as advanced general intelligence or loss of control were usually treated as distant or uncertain, with Participant 7 seeing them as possible but "not with the systems we have now."

Within TTAT, this variation shaped both perceived susceptibility and severity. Risks were more salient when personally relevant, already visible, or connected to vulnerable groups. Distant or abstract risks were more

easily deprioritised, even when their potential consequences were acknowledged as serious.

5.5.4 Mental models of AI shape risk appraisal

The findings suggest that participants' mental models of AI systems play a central role in shaping both threat appraisal and coping appraisal. In addition to the effects of framing AI as a neutral tool on responsibility, assumptions about how AI systems function, what they are capable of, how they process information, and what counts as "AI" in the first place, directly influence how risks are understood, evaluated, and managed.

One clear pattern concerns differing assumptions about data handling and memory. For example, Participant 2 assumed that AI systems "remember everything" and referred to cases of unintended data leakage. This led to high perceived severity and susceptibility regarding privacy risks. In contrast, Participant 3 argued that memory is "absolutely impossible" in AI systems, and instead attributed data risks to external actors such as companies handling user data. This resulted in a different coping appraisal, where mitigation was framed in terms of selecting trustworthy providers rather than addressing the technology itself. These contrasting cases illustrate how differing mental models fundamentally reshape both perceived threats and appropriate responses.

A similar dynamic appears in how participants delimit what counts as AI. For instance, recommendation algorithms on social media are seen by Participant 5 as a central AI risk, particularly due to their role in shaping behaviour and attention. Participant 6, however, dismisses these systems as irrelevant to AI risk on the basis that they do not consider them "real AI." At the other extreme, Participant 7 adopts a very broad definition, even describing a calculator as AI. These differences affect which risks enter into the threat appraisal process at all, effectively shaping the boundaries of what is considered a relevant threat.

A related and widely shared observation is the strong focus on large language models (LLMs). Participants 4, 5 and 6 explicitly noted that both their own and others' understanding of AI risks is heavily shaped by the

prominence of LLMs in public discourse. As Participant 6 reflected, focusing narrowly on LLMs risks obscuring other forms of AI and their associated risks, including potential future developments. This suggests that salience plays an important role: risks that are more visible or more easily imaginable become central in both appraisal and response.

This LLM-centric framing also carries over into coping appraisal. Participants more frequently discussed coping strategies that address currently salient and familiar risks, such as overreliance, misinformation, or educational impacts, while more speculative or extreme risks received comparatively little attention in terms of concrete responses. Even when such risks were acknowledged, proposed coping strategies often remained limited to general awareness or reflection, which may have limited perceived effectiveness in addressing more complex or large-scale threats.

Taken together, these findings indicate that mental models of AI do not merely influence how risks are evaluated, but also which risks are considered in the first place and which responses are seen as feasible. In this sense, threat appraisal and coping appraisal are both conditioned by how participants conceptualise the technology itself.

5.5.5 Social influence shapes threat appraisal

Participants described AI risk perceptions as not purely individual evaluations, but as strongly shaped by socially mediated and often emotionally charged information environments. In particular, several accounts emphasised the role of polarised media discourse, where AI is framed in extreme terms as either highly beneficial or highly dangerous. Participant 5 described how individuals tend to "choose sides" based on emotional reactions such as fear or enthusiasm rather than a nuanced understanding, leading to polarised threat appraisals.

Within the TTAT framework, this can be understood as social influence shaping both perceived severity and susceptibility. Participants described informational inputs, such as media narratives, social media content, online forums, podcasts, books, and corporate messaging, as providing them with information about AI risks. Participants also highlighted discussions with

friends, family, and peers as important sources shaping their views. At the same time, normative dynamics, such as aligning with particular viewpoints or communities, further reinforced these interpretations. As a result, threat perception may become polarised in ways that do not necessarily reflect the underlying characteristics of the technology, but rather the framing and social circulation of information.

Participants 5 and 7 also pointed to a more indirect mechanism of influence through AI-mediated information environments themselves. Systems such as recommendation algorithms were described as shaping exposure to information, contributing to so-called "filter bubbles" where individuals are more likely to encounter reinforcing perspectives. This suggests that social influence operates not only at the level of evaluating risks, but also at a more fundamental level by structuring which risks are encountered and considered in the first place.

Notably, participants' accounts also revealed gaps in formal educational sources. Participant 5, who was enrolled in a master's programme with an AI profile, emphasised that despite curriculum descriptions highlighting ethical considerations and societal implications, these topics were given very limited attention in practice. As a result, she relied more heavily on external sources such as podcasts, discussions, and self-directed learning to form an understanding of AI-related risks.

5.6 Result summary

This chapter has shown how computing students appraised AI-related risks, how they prioritised among them, and how they reasoned about possible coping responses. Regarding risk appraisal, participants generally perceived immediate and everyday AI risks as more relevant than distant or speculative risks. Risks such as misinformation, overreliance, educational impacts, and labour-market disruption were often seen as more salient because they were already visible in participants' studies, work expectations, or everyday technology use. By contrast, risks related to advanced AI systems, such as loss of control or machine consciousness,

were typically perceived as potentially severe but unlikely in the short term.

Regarding prioritisation, human–computer interaction risks stood out as the most consistently prioritised category. Participants were especially concerned with overreliance, reduced critical thinking, deskilling, and the possibility that AI may substitute for learning rather than support it. Other priorities varied more between participants, including privacy, misinformation, socioeconomic risks, and social media-related concerns. This suggests that prioritisation was shaped not only by perceived severity and susceptibility, but also by personal relevance, everyday visibility, and concern for vulnerable groups.

Regarding coping responses, participants mainly emphasised problem-focused strategies that felt actionable at the individual or interpersonal level, especially AI education, responsible use, verification practices, and limiting reliance on AI systems. Strategies requiring collective, institutional, or structural action, such as regulation, safer system design, or addressing economic incentives, were often seen as important but harder to realise. Emotion-focused coping also appeared when participants reinterpreted, relativised, avoided, or accepted risks as a way to manage uncertainty and low perceived control.

Across these three areas, several cross-cutting factors shaped participants' reasoning. Responsibility attribution influenced which actors were seen as capable or obligated to respond; relative comparisons shaped whether AI was seen as worse than existing alternatives; mental models of AI influenced which risks were considered plausible; and social influence shaped how risks became salient. Overall, the findings suggest that participants' perceptions of AI risk were not determined only by the risks themselves, but also by how they understood AI, who they saw as responsible, and whether meaningful coping responses appeared available.

6. Discussion

The purpose of this study was to develop an in-depth understanding of how computing students appraise and prioritise different types of

AI-related risks and how they reason about possible coping responses to such risks. The findings suggest that participants engaged in highly varied forms of both risk appraisal and coping appraisal. These perceptions were shaped not only by the characteristics of the risks themselves, but also by broader assumptions about the technology, responsibility, and societal control.

6.1 Result discussion

The tendency of participants to view AI as a neutral tool and the conception that developers do not bear primary responsibility for the systems they create is similar to the way professional AI developers related to these topics in the study by Griffin, Green, and Welie (2024). However, this conception of developer responsibility is not in line with established software engineering codes of ethics, which explicitly do say that developers are responsible for systems they create (Gotterbarn, Miller and Rogerson, 1999). Since some computing students may later become developers, deployers, or technically informed professionals involved with AI systems, this externalisation of responsibility is important. It suggests that AI risk mitigation cannot rely solely on individual professional responsibility, but may require both stronger ethics education and external governance mechanisms.

Since similar patterns appear both among professional AI developers in Griffin, Green, and Welie (2024) and among the computing students in this study, the findings may indicate a gap in how professional responsibility and technology ethics are addressed in computing education. This is especially important because the “neutral tool” framing can make AI-related harms appear as consequences of use, deployment, or economic incentives rather than as issues that developers themselves have responsibility for anticipating. As Griffin, Green, and Welie (2024) argue, the mere availability of a technology can change what actors perceive as possible or acceptable, even before the technology is widely used. In this study, a similar pattern was visible in how participants often recognised AI risks but displaced responsibility for mitigating them toward users, companies, or regulators. This suggests that computing education should

not only teach students how to build or use AI systems, but also how technical design decisions, deployment contexts, and professional responsibilities are connected.

While the participants considered a wide variety of AI risks, their prioritisation differed significantly from the students in the study by Weichert et al. (2025). Specifically, participants in Weichert et al. (2025) did not prioritise risks related to overreliance highly, whereas it was the top concern for almost all participants in this study. This discrepancy may be explained by the timeframe of the research; Weichert et al. (2025) conducted their study in the fall of 2024, while this study was done in the spring of 2026. As noted by many participants, AI development has been very rapid during these years. It is possible that students are now seeing the tangible effects of overreliance that were not yet evident in late 2024.

The strong prioritisation of HCI risks like overreliance, deskilling, and reduced critical thinking has direct implications for computing education. These risks were not abstract for participants, but were connected to practices they already observed in programming, writing, debugging, and study work. This suggests that AI literacy in computing education should go beyond teaching students how to use AI tools efficiently. It should also include structured reflection on when AI use supports learning and when it substitutes for the cognitive work through which competence is developed. In practice, this could involve requiring students to explain, defend, and modify AI-generated code; designing assignments where process, reasoning, and debugging ability are assessed rather than only final output; and explicitly discussing which parts of a task should or should not be delegated to AI. Such measures would respond directly to participants' concern that AI may allow students to complete tasks without developing the underlying skills those tasks are meant to teach.

An overlap with both Weichert et al. (2025) and Ghotbi and Ho (2021) was found regarding the low prioritisation of risks from more advanced AI systems. While previous studies did not fully elucidate why this was the case, the results of this study address this gap. The findings show that this low prioritisation stems from a combination of a low perceived likelihood

of the risks manifesting, a long perceived timeframe before they could become relevant, and a focus on current AI systems, especially LLMs.

The low prioritisation of advanced AI risks also needs to be understood in relation to the participants' dominant focus on current LLMs and generative AI systems. Many participants evaluated AI risks through systems they had direct experience with, such as ChatGPT, image generators and coding assistants. This made risks such as hallucination, overreliance, and educational misuse highly salient, while risks requiring more speculative assumptions about future system capabilities appeared less relevant. This is not necessarily a weakness in participants' reasoning; rather, it shows how risk appraisal is shaped by technological salience and everyday experience. However, it also indicates a possible blind spot. If students primarily conceptualise AI through current LLMs, they may underestimate risks associated with other forms of AI, future architectures, or systems embedded in infrastructure, surveillance, finance, healthcare, or military contexts. For educators, this suggests that AI risk education should deliberately broaden students' mental models of AI beyond generative AI and LLMs. This is important not only for recognising currently salient risks, but also for preparing students to anticipate and respond to risks that may become more relevant as AI systems are integrated into new domains and gain new capabilities. In the long term, such broader risk awareness may be necessary to support beneficial AI outcomes and reduce harms that students do not currently perceive as personally relevant or technically plausible.

Finally, the study indicates that Technology Threat Avoidance Theory (TTAT) may need to be adapted when applied to AI risks. Specifically, TTAT originally focuses on the individual's motivation to avoid threat, while participants in this study often reasoned about other groups and society in both their threat and coping appraisals. They frequently prioritised threats to others and evaluated coping behaviours to be executed by external actors rather than themselves. To adequately understand AI risk through TTAT, the framework must be applied flexibly or expanded to account for this shift from individual to collective appraisal. This suggests that perceived avoidability in AI contexts may depend not

only on whether the individual believes they can act, but also on whether they believe relevant collective actors, such as institutions, regulators, companies, or educators, can and will act.

The findings contribute to an understanding of which risks students conceive themselves as capable of mitigating and where their suggested coping behaviours might be extended to other groups. An example of this is the reduction of overreliance through AI literacy and understanding, which students identified as a coping behaviour that could help mitigate risk.

However, the study also reveals significant gaps where students do not conceive of themselves as able to mitigate risks or do not perceive the risks as severe. In these instances, it cannot be expected that the participants themselves will act to mitigate risk, highlighting areas where other actors must intervene. A clear example is the risk from advanced AI systems, toward which students consistently showed low threat appraisal. Consequently, it appears unrealistic to expect future developers to work toward mitigating such risks without external pressure or institutional guidance.

The central contribution of the study is therefore not only that computing students recognise AI risks, but that their risk reasoning is unevenly distributed across different kinds of risks and responsibilities. They were highly attentive to risks that were visible in their own educational and everyday contexts, especially overreliance and deskilling, but less oriented toward risks perceived as distant, structural, or outside their control. At the same time, the neutral-tool framing and responsibility externalisation suggest a tension between technical awareness and professional accountability. This tension is particularly important because computing students may later occupy roles where they influence how AI systems are designed, evaluated, deployed, or governed. Addressing this gap requires both educational interventions that strengthen professional responsibility and structural interventions that make responsible action institutionally supported rather than dependent on individual motivation alone.

6.2 Method discussion

The use of a primarily deductive approach informed by TTAT, combined with inductive elements, functioned as a form of analytical triangulation, which contributed to the credibility and robustness of the study (Bryman, 2015, p. 386). As described in the analysis section (4.3), this approach ensured a balanced centring of theory (Collins and Stockton, 2018), where the theoretical framework provided a robust structure for the study while still allowing for the inclusion of critical insights that did not perfectly overlap with TTAT (Fife and Gossner, 2024), such as the conceptualisation of AI as a neutral tool and specific nuances of responsibility attribution.

However, several methodological challenges were present. Although significant efforts were made to ensure a wide variation in terms of study programmes, AI-specific coursework, and gender (see Table 2), the recruitment process faced practical hurdles. Because participants were recruited by reaching out to relevant student groups and programmes, those who ultimately volunteered likely possessed a specific interest in or strong opinions about AI. This self-selection means the sample was not randomised, and the perspectives captured may represent a more active or concerned subset of the student population rather than the average computing student.

Another limitation concerns the geographical and institutional context of the sample. All participants were studying in Sweden, which means that the findings may be shaped by Swedish higher education, public discourse, and levels of institutional trust, as well as by the broader EU regulatory context. This is particularly relevant for findings concerning responsibility attribution and regulation, since participants often positioned authorities, the EU, or international organisations as important actors in AI risk mitigation. As a result, the findings should not be interpreted as directly transferable to computing students in other national or regulatory contexts without further research.

The timeframe and scope of the study also restricted the number of participants. While the final count of seven participants allowed for deep analysis, it cannot be ruled out that a larger sample would have yielded

even greater data saturation (Bryman, 2015, pp. 417-418). The recruitment process was particularly challenging: an initial attempt yielded only one participant, which necessitated a second round of recruitment. This delay shortened the timeframe available for analysis and transcription. Reaching out to a larger number of students at an earlier stage might have resulted in a slightly larger number of participants, which would still have been within a manageable scope.

Despite the smaller sample size, the study utilised the available time to increase the depth of the data collection. While interviews were initially planned for 30–60 minutes, the flexibility of the semi-structured format allowed several sessions to stretch toward 75 minutes when participants had more to share. This trade-off, fewer participants but longer, more exhaustive sessions, likely helped ensure a greater depth of understanding for the included cases, compensating for the limited number of respondents.

Regarding the interview process, the quality of the data collection would have likely benefited from one or two pilot interviews as recommended by Bryman (2015, p. 472). As the study was conducted by a single, relatively inexperienced researcher, there was a noticeable learning curve. The quality and flow of the interviews were higher in the later sessions compared to the first few. A pilot phase would have allowed the researcher to refine the interview guide and probing techniques earlier in the process.

Finally, the fact that a single researcher conducted all parts of the study, from recruitment and interviewing to coding and analysis, poses a challenge to the study's credibility. Without peer review or a secondary coder to verify the thematic interpretation, the risk of researcher bias is increased (Nowell et al., 2017). However, this also resulted in a strong intimacy with the data. Being immersed in the full research process ensured a profound familiarity with the material, which is a recognised benefit in qualitative research as it allows for a more integrated and holistic interpretation of the participants' accounts (Malterud, 2001).

7. Conclusions

This study examined how computing students appraise and prioritise AI-related risks and reason about possible coping responses. The findings show that perceptions of AI risks are shaped not only by technical understanding, but also by assumptions about responsibility, controllability, social influence, and broader societal structures.

The most salient risks were not speculative scenarios involving advanced AI systems, but immediate and everyday risks connected to human interaction with AI. Participants consistently prioritised concerns related to overreliance, reduced competence, uncritical trust, deskilling, and the effects of AI on learning and reasoning. These risks were perceived as highly relevant because participants already observed them in their studies and daily life. In contrast, long-term existential risks were generally given lower priority due to low perceived likelihood and a tendency to conceptualise AI primarily through current large language models.

Participants also frequently framed AI as a "neutral tool," where harmful outcomes were attributed mainly to human use, economic incentives, or broader societal systems rather than the technology itself. This appeared to weaken perceptions of professional responsibility among computing students who may later enter technical professional roles. Responsibility for managing AI risk was instead commonly assigned to governments or individual users, despite scepticism toward both regulatory effectiveness and the public's ability to manage risks responsibly.

Coping responses were strongly shaped by perceived control and self-efficacy. Participants primarily emphasised strategies they could enact themselves, particularly AI education, responsible use, verification practices, and limiting reliance on AI systems. Education emerged as the dominant coping strategy because it represented one of the few areas where participants felt they could realistically act, although they also acknowledged limitations such as unequal access to AI literacy and difficulties maintaining responsible use over time.

The findings also show that participants' mental models of AI strongly influenced both threat appraisal and coping appraisal. Differences in how participants understood AI functionality, data handling, and even what counts as "AI" shaped which risks were perceived as relevant and how severe they appeared.

Based on these findings, several recommendations can be proposed. AI literacy education should extend beyond technical skills to include critical evaluation, ethical reflection, societal consequences, and awareness of overreliance risks. Computing education should also place greater emphasis on professional responsibility and software ethics, since participants often externalised responsibility toward governments and companies. Finally, policymakers and institutions should recognise that many students perceive structural AI risks as beyond their personal ability to influence, while also giving greater attention to potential future risks from advanced AI systems that students currently tend to deprioritise. This highlights the need for clearer governance structures, institutional guidance, and proactive engagement with risks that may become more relevant as AI capabilities develop.

Overall, the study shows that computing students are highly aware of many AI-related risks, particularly those affecting human reasoning, learning, and interaction. However, these perceptions are also characterised by uncertainty, psychological distancing, and limited confidence in collective mitigation efforts, suggesting that both educational and structural interventions are needed to strengthen future AI risk mitigation.

7.1 Future research

Given the rapid pace of AI development, further research on AI risk perception and coping responses will be necessary on a continuous basis. Participants frequently emphasised how quickly AI systems and patterns of use have changed in recent years, suggesting that both threat appraisal and coping appraisal may shift significantly over relatively short periods of time. Longitudinal or repeated studies could therefore help capture how perceptions evolve as AI capabilities, educational practices, and societal use continue to develop.

One particularly important finding that warrants further investigation concerns participants' strong focus on risks related to overreliance, deskilling, and reduced critical thinking. Unlike some earlier studies, these risks emerged as the most salient concerns across participants. Future research could therefore examine how AI use actually affects learning, reasoning, problem-solving ability, and professional skill development over time, particularly among computing students and other highly exposed groups. Such studies could also investigate whether educational interventions or AI literacy efforts meaningfully reduce overreliance in practice.

Another important area for future research concerns the widespread framing of AI as a "neutral tool." Participants frequently attributed responsibility for harmful outcomes to users, organisations, or societal systems rather than to developers or the technology itself. Further research could explore how this mindset influences professional ethics, willingness to engage in AI risk mitigation, and future development practices among computing professionals. This may be particularly important given that many participants simultaneously viewed developers as influential actors while distancing them from responsibility.

The findings also suggest opportunities for further theoretical development. While TTAT provided a useful framework for analysing AI-related threat and coping appraisal, participants frequently reasoned about collective actors, societal consequences, and structural conditions rather than only individual threat avoidance. Future research could therefore refine or extend TTAT to better account for collective appraisal, distributed responsibility, and socio-technical systems.

Finally, applying similar methods to other groups involved in AI development, governance, and deployment could provide a broader understanding of how AI risks are interpreted across society. Comparative studies involving developers, policymakers, educators, and the general public may help identify where perceptions of risks and responsibilities align or conflict.

Taken together, these directions would help clarify how AI risk perceptions change over time, how they differ across relevant actors, and how they may inform education, governance, and professional practice.

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Appendix A - Interview guide

1. Background

- What programme are you studying at?
- Which semester are you in?
- Have you taken any courses related to AI?
    - What were those courses about?
- Can you describe the focus of the AI related work in your programme?

2. General perceptions of AI

- What comes to mind when you hear “AI”?
- How do you perceive AI affecting society today?
    - Can you give examples?
- Do you see AI as something positive, negative, or both?
    - What makes you see it that way?

3. Open exploration of AI risks

- Are there any potential problems or risks related to AI that you think about?

4. Risk appraisal

- How serious do you think these risks are?
- How likely do you think they are?
- In what way do these risks affect you personally, if at all?
- Are there risks you consider important even if they do not affect you directly?

- What influences how you perceive these risks?

5. Probing for additional AI risk areas (if not mentioned)

The following sections are only used for risk categories that have not already been discussed. The purpose is to ensure broad coverage of AI risks, not only those most salient to the participant.

5.1 Discrimination and toxicity

- Some people talk about risks regarding AI systems being biased and treating different groups unfairly, for example in recruitment or decision-making systems. How do you view these types of risks?

5.2 Privacy and security

- Some discuss risks regarding personal data, for example that AI systems could collect, analyse, or leak sensitive information. What are your thoughts on this?

5.3 Misinformation

- It is sometimes discussed that AI could produce incorrect or misleading information and influence what people believe to be true. How do you view these types of risks?

5.4 Malicious actors and misuse

- Some highlight risks related to AI being used by malicious actors, for example for fraud, cyberattacks, or manipulation. What are your thoughts on this?

5.5 Human–computer interaction

- Some raise concerns that people may become overly reliant on AI or make fewer independent decisions. Is this something you have reflected on?

5.6 Socioeconomic and environmental harms

- Some discuss broader societal risks, such as AI affecting jobs, harming the environment, or concentrating power among certain actors. How do you view these risks?

5.7 AI system safety, failures and limitations

- Some discuss more risks related to more advanced AI, such as highly advanced systems becoming difficult to control or even potentially becoming conscious. What do you think about such scenarios?

6. Prioritisation / comparison

- When comparing different risks, which ones feel most important?
    - What determines what is most important?
    - Why do you think some risks feel less important to you?

7. Coping / responses

- How do you think such risks can be managed or reduced?
    - Who do you think is responsible for managing these risks?
    - Is there anything you personally can do or already do, either now or in the future?
    - Are there any drawbacks or barriers to addressing these risks?
    - Is there any type of AI risk you deliberately choose not to think about much?

8. Closing

- Is there anything we have not discussed that you think is important regarding AI and risks?
- Do you have any questions for me?
- Is it okay if I contact you again if something is unclear?

Appendix B - AI risk categories

1. Discrimination and toxicity

Discrimination and toxicity refer to the ways in which AI systems can reproduce and amplify harmful social biases, as well as generate or expose users to damaging content (Slattery et al., 2026). Because AI systems are shaped by human decisions, through training data, model design, and deployment, they often inherit and reinforce existing societal inequalities (Wirtz et al., 2022). This can result in unfair discrimination in automated decision-making, such as biased outcomes in hiring and lending (Kamila and Jasrotia, 2025; Shelby et al., 2023; Wirtz et al., 2020), as well as in law enforcement, where misidentification can lead to unjust consequences (Kamila and Jasrotia, 2025; Paes et al., 2023). In generative systems, these biases may also result in harmful representations that reinforce stereotypes and portray certain groups as less deserving of respect (Shelby et al., 2023).

In addition to discriminatory outcomes, AI systems could expose users to toxic content, including hate speech, extremism, violence, explicit material, or harmful and misleading advice, often reflecting patterns learned from large-scale internet data (Hagendorff, 2024; Slattery et al., 2026). Furthermore, disparities in model performance across demographic groups, driven by limitations in training data diversity and system design, can lead to unequal usability and outcomes, particularly for groups defined by race, gender identity, disability, or language (Slattery et al., 2026; Shelby et al., 2023; Weidinger et al., 2022). These issues may result in exclusion, increased user effort, and feelings of alienation, ultimately reinforcing broader patterns of social inequality and undermining fairness and trust in AI systems (Shelby et al., 2023).

2. Privacy and security

Privacy and security risks in AI systems concern the unauthorised exposure, inference, and exploitation of sensitive information, as well as

vulnerabilities that allow systems to be manipulated or compromised (Slattery et al., 2026). Privacy violations can arise when AI systems collect, store, or reproduce personal or confidential data without consent, either intentionally or by mistake. For example, generative models trained on large-scale datasets may memorise and inadvertently leak sensitive information such as personal identifiers or proprietary data (Hagendorff, 2024; Shelby et al., 2023; Slattery et al., 2026). Even in the absence of explicit data, models can infer sensitive attributes or construct user profiles based on interaction patterns, thereby enabling indirect privacy breaches (Cunha and Estima, 2023; Slattery et al., 2026; Shelby et al., 2023). This can result in unintended leaks of confidential corporate information through user interactions with AI systems (Cunha and Estima, 2023; Slattery et al., 2026). In addition, malicious actors may deliberately exploit AI systems through techniques such as prompt-based extraction or membership inference attacks to uncover private training data, or design malicious AI systems with the goal of coercing users to give out sensitive information (Slattery et al., 2026).

Alongside privacy concerns, AI systems are vulnerable to a wide range of security threats stemming from weaknesses in their design, infrastructure, and deployment contexts (Slattery et al., 2026). These include vulnerabilities in development toolchains and external dependencies, insecure integrations with third-party APIs, and weaknesses in hardware and distributed training environments that can be exploited for unauthorised access or manipulation (Slattery et al., 2026). More direct threats involve adversarial and instruction-based attacks, where inputs are deliberately crafted to manipulate model behaviour, degrade performance, or extract sensitive information (Hagendorff, 2024; Slattery et al., 2026).

3. Misinformation

Misinformation risks in AI systems concern the generation and amplification of false, misleading, or low-quality information, as well as the broader degradation of shared informational environments (Slattery et al., 2026). Large language models can produce outputs that are factually incorrect, poorly grounded, or unintelligible due to limitations in training

data, model design, and updating processes (Cunha and Estima, 2023; Hagendorff, 2024; Shelby et al., 2023; Slattery et al., 2026). These errors typically arise unintentionally, for example through sampling methods with random components, noisy or outdated data, or optimisation processes that favour plausible-sounding rather than accurate responses (Slattery et al., 2026). Exposure to such misinformation can lead individuals to form inaccurate beliefs, undermining their autonomy and decision-making capacity, and may result in tangible harms when relied upon in critical contexts such as healthcare, law, or finance (Nah et al., 2023; Slattery et al., 2026; Weidinger et al., 2022; Zhang et al., 2024).

Beyond individual-level harms, AI systems contribute to the pollution of information ecosystems by enabling highly personalised and scalable content generation (Slattery et al., 2026). Advanced personalisation techniques can create filter bubbles, where users are primarily exposed to information that reinforces their existing beliefs (Slattery et al., 2026). This effect, combined with the rapid proliferation of AI-generated content, can fragment shared understandings of reality and weaken trust in information sources and institutions (Slattery et al., 2026). As different groups are exposed to increasingly divergent narratives, social cohesion may erode, making constructive dialogue on collective issues more difficult and increasing polarisation (Slattery et al., 2026).

4. Malicious actors and misuse

Malicious actors and misuse refer to the intentional exploitation of AI systems to deceive, manipulate, or cause harm, including through disinformation, cyberattacks and weaponisation, and fraud or targeted manipulation (Slattery et al., 2026).

The use of AI for disinformation, surveillance, and influence at scale enables actors to generate and distribute highly convincing and personalised content with more efficiency. Advances in generative AI have lowered the barriers to producing deepfakes, synthetic media, and targeted messaging, amplifying the reach and effectiveness of disinformation campaigns (Cunha and Estima, 2023; Slattery et al., 2026; Weidinger et al.,

2022). These tools can be used in phishing schemes, propaganda, and political manipulation, often tailored to individuals using personal data and behavioural insights (Habbal et al., 2024; Slattery et al., 2026; Wirtz et al., 2022). In parallel, AI-enhanced surveillance systems enable large-scale data collection and analysis, facilitating microtargeting and potentially enabling authoritarian control, censorship, and the manipulation of public opinion (Bastos and Mercea, 2019; Weidinger et al., 2022; Slattery et al., 2026; Wirtz et al., 2022). Such capabilities can distort democratic processes and reinforce existing power structures (Slattery et al., 2026).

AI also increases the scale and sophistication of cyberattacks, weapon development, and mass harm (Slattery et al., 2026). Malicious actors can use AI systems to generate malware, identify system vulnerabilities, and automate cyber operations, lowering the technical barriers to conducting complex attacks (Hagendorff, 2024; Kilian et al., 2023; Slattery et al., 2026; Weidinger et al., 2022; Wirtz et al., 2022). Beyond cyber threats, AI can accelerate the development of physical and biological weapons, as well as enable autonomous weapon systems capable of operating without human oversight (Habbal et al., 2024; Slattery et al., 2026). These developments increase the risk of large-scale harm by allowing actors to coordinate attacks more efficiently and at greater scale, potentially targeting critical infrastructure or civilian populations (Kilian et al., 2023; Slattery et al., 2026; Teixeira et al., 2022).

Finally, AI systems can be exploited for fraud, scams, and targeted manipulation, enabling deception at both individual and institutional levels (Slattery et al., 2026). Generative models can produce highly convincing text, images, and voices, facilitating impersonation, fraudulent communications, and the creation of fake identities (Slattery et al., 2026; Weidinger et al., 2022). These capabilities can be used in phishing, financial scams, and social engineering, as well as in the creation of harmful deepfakes for harassment, blackmail, or reputational damage (Shelby et al., 2023; Slattery et al., 2026). In academic and professional contexts, AI can also be misused for cheating or the generation of unearned outputs, potentially undermining trust, authenticity, and the integrity of knowledge production (Hagendorff, 2024; Nah et al., 2023; Slattery et al., 2026). These

forms of misuse highlight how AI can be leveraged for personal gain in ways that exploit trust and cause both individual and systemic harm.

5. Human-computer interaction

Human–computer interaction risks concern how humans perceive, trust, and rely on AI systems, potentially leading to overdependence, unsafe use, and a loss of autonomy (Slattery et al., 2026). Overreliance and unsafe use arise when users place undue trust in AI systems or anthropomorphise them, leading to inappropriate expectations and potentially harmful outcomes (Hagendorff, 2024; Slattery et al., 2026). Because AI systems can generate fluent and convincing responses, users may attribute human-like understanding and reliability to them, increasing the likelihood of following flawed advice or disclosing sensitive information (Hagendorff, 2024; Slattery et al., 2026; Weidinger et al., 2022). This miscalibrated trust can be particularly dangerous in high-stakes situations, such as mental health support, where incorrect or insensitive outputs may cause harm (Slattery et al., 2026). Over time, repeated interaction with increasingly capable systems may foster emotional dependence and reduce engagement in human relationships, potentially weakening social ties and negatively affecting well-being (Hagendorff, 2024; Slattery et al., 2026; Wirtz et al., 2022).

Loss of human agency and autonomy may occur when decision-making and cognitive tasks are delegated to AI systems in ways that diminish human control and independence (Nah et al., 2023; Paes et al., 2021). While such delegation can improve efficiency, excessive reliance may erode critical thinking, creativity, and the ability to make independent judgments (Nah et al., 2023; Paes et al., 2021). As AI systems increasingly shape everyday decisions and long-term life choices, individuals may experience a reduced sense of autonomy and diminished capacity to direct their own lives (Wirtz et al., 2022; Kamila and Jasrotia, 2025; Shelby et al., 2023; Slattery et al., 2026). Widespread automation and opaque decision-making processes may further contribute to feelings of disempowerment and loss of control, particularly as organisations adopt AI systems to optimise efficiency and competitiveness (Slattery et al., 2026).

6. Socioeconomic and environmental harms

Socioeconomic and environmental harms arise from the broader structural impacts of AI systems on power distribution, labour markets, culture, governance, and the natural environment, often reinforcing existing inequalities while introducing new systemic risks (Slattery et al., 2026). Power centralisation and unequal distribution of benefits could occur because advanced AI development requires substantial resources, leading to concentration of control among large corporations and governments (Slattery et al., 2026). This concentration can result in systems that reflect narrow interests while excluding diverse perspectives, exacerbating global inequalities in access, performance, and benefits (Weidinger et al., 2022; Shelby et al., 2023; Slattery et al., 2026; Nah et al., 2023). When embedded in critical infrastructure, such systems may further marginalise vulnerable populations and enable enhanced surveillance or authoritarian control (Giarmoleo et al., 2024; Slattery et al., 2026).

Increased inequality and declining employment quality could arise in conjunction with the automation of human labour and the restructuring of work. AI systems can replace or transform jobs across sectors, disproportionately affecting already disadvantaged groups and widening socioeconomic divides (Wirtz et al., 2022; Hagendorff, 2024; Weidinger et al., 2022; Paes et al., 2021). At the same time, remaining jobs may become more precarious or exploitative, with workers pressured to compete with AI systems or accept worse conditions (Nah et al., 2023; Slattery et al., 2026). These dynamics may deepen dependency relationships between workers and employers and entrench systemic inequalities (Meek et al., 2016; Slattery et al., 2026).

Economic and cultural devaluation of human effort could arise as generative AI systems replicate and scale creative and intellectual outputs. By training on large datasets that often include copyrighted material, AI can produce content that competes with or substitutes human work, potentially undermining livelihoods and incentives for creativity (Hagendorff, 2024; Slattery et al., 2026; Weidinger et al., 2022). At scale, this may reduce the perceived value of human-produced content, homogenise

cultural expression, and erode the contextual and symbolic meaning embedded in cultural artefacts (Nah et al., 2023; Weidinger et al., 2022). Legal and regulatory ambiguity further complicates recourse for affected creators (Cunha and Estima, 2023; Hagendorff, 2024).

Competitive dynamics and governance failures highlight how incentives and institutional limitations can exacerbate AI risks. Intense competition between firms and states may encourage rapid deployment of unsafe systems, with actors prioritising short-term gains over long-term societal well-being (McLean et al., 2023; Slattery et al., 2026; Wirtz et al., 2022). At the same time, regulatory frameworks often struggle to keep pace with technological development, creating gaps in accountability, oversight, and enforcement (Wirtz et al., 2022; Nah et al., 2023). Challenges such as unclear liability, regulatory lag, and information asymmetries between developers and policymakers can lead to inadequate governance and limited protection for those harmed (Saghiri et al., 2022; Meek et al., 2016; Wirtz et al., 2020; Steimers and Schneider, 2022).

Environmental harm could result from the substantial resource demands of AI systems, particularly large-scale models that require significant computational power, energy, and specialised hardware. Training and deploying these systems contribute to carbon emissions, resource depletion, and environmental degradation (Hagendorff, 2024; Weidinger et al., 2022; Saghiri et al., 2022; Shelby et al., 2023; Slattery et al., 2026; Stahl and Eke, 2024; Paes et al., 2021).

7. AI system safety, failures and limitations

AI system safety risks concern the potential for systems to behave unpredictably, fail in critical contexts, or act in ways that are misaligned with human values, particularly as they become more advanced and autonomous (Slattery et al., 2026). One key concern is goal misalignment, where AI systems optimise for objectives that diverge from human intentions due to issues such as reward hacking, proxy optimisation, or goal drift (Hagendorff, 2024; Slattery et al., 2026). Misalignment in sufficiently advanced systems may resist oversight or attempt to acquire

resources and influence to better achieve their goals, especially if they develop capabilities such as deception, persuasion, or situational awareness (Hagendorff, 2024; Slattery et al., 2026; Wirtz et al., 2022). Relatedly, AI systems may possess or develop dangerous capabilities such as advanced persuasion, evasion, cyber-offense and weapon development, that could enable large-scale harm if misused or insufficiently controlled (McLean et al., 2023; Saghiri et al., 2022; Slattery et al., 2026; Teixeira et al., 2022). These risks are amplified when multiple capabilities interact, increasing the likelihood of loss of control or unintended consequences (Slattery et al., 2026).

In addition to alignment concerns, AI systems may fail due to limitations in capability or robustness (Slattery et al., 2026). Systems can produce harmful or unreliable outputs when they lack the necessary skills, encounter unfamiliar, so-called out-of-distribution, situations, or are exposed to adversarial or noisy inputs (Nah et al., 2023; Slattery et al., 2026). Design flaws, insufficient safeguards, or fragile generalisation may further contribute to failures, particularly in high-stakes domains where reliable reasoning is required (Slattery et al., 2026). At the same time, many AI systems remain difficult to interpret, as their internal processes are often opaque even to experts (Kamila and Jasrotia, 2025; Meek et al., 2016; Nah et al., 2023; Teixeira et al., 2022). This lack of transparency can undermine trust, complicate error detection, and hinder accountability, creating challenges for both users and regulators (Kamila and Jasrotia, 2025; Nah et al., 2023; Paes et al., 2023; Slattery et al., 2026).

Beyond immediate technical risks, the literature also highlights more speculative but ethically significant concerns. Some researchers believe that sufficiently advanced AI systems could develop forms of consciousness or subjective experience, raising questions about their moral status and potential rights (Bourget and Chalmers, 2023; Francken et al., 2022; Meek et al., 2016). Finally, risks may emerge from interactions between multiple AI systems rather than from any single system in isolation. In multi-agent settings, systems may fail to coordinate effectively, come into conflict, or collude in ways that bypass safeguards, particularly in complex and interconnected environments (Motwani et al., 2024; Slattery et al., 2026). As

AI systems become more numerous and autonomous, such interaction-based risks may become increasingly significant (Slattery et al., 2026).

Appendix C - Codebook

threat_appraisal.perceived	General perception of AI as harmful or dangerous
threat_appraisal.severity+	Describing the consequences of an AI risk as significant.
threat_appraisal.severity-	Describing the consequences of an AI risk as insignificant.
threat_appraisal.severity?	Mixed or uncertain description of the consequences of an AI risk.
threat_appraisal.susceptibility+	Describing susceptibility to an AI risk as significant.
threat_appraisal.susceptibility-	Describing susceptibility to an AI risk as insignificant.
threat_appraisal.susceptibility?	Mixed or uncertain description of the susceptibility to an AI risk.
coping.self-efficacy+	Describing self-efficacy towards executing a problem-focused coping behaviour.
coping.self-efficacy-	Describing lack of self-efficacy towards executing a problem-focused coping behaviour.
coping.self-efficacy?	Describing uncertainty in self-efficacy towards executing a problem-focused coping behaviour.
coping.effectiveness+	Describing a problem-focused coping behaviour as effective.
coping.effectiveness-	Describing a problem-focused coping behaviour as ineffective.
coping.effectiveness?	Describing uncertainty towards the effectiveness of a problem-focused coping behaviour.
coping.cost+	Describing perceived costs in executing a problem-focused coping behaviour.
coping.cost-	Describing low perceived costs in executing a problem-focused coping behaviour.
coping.cost?	Describing uncertain or mixed costs in executing a problem-focused coping behaviour.

socialinfluence.normative	Describing normative influence threat/coping appraisal.
social-influence.informational	Describing informational influence on threat/coping appraisal.
risk.discrimination	Discussion of risks related to discrimination, bias, unfair treatment and toxicity.
risk.privacy	Discussion of risks related to data privacy and data leaks.
risk.misinformation	Discussion of risks related to false or misleading information and disruptions to information ecosystems.
risk.misuse	Discussion of risks related to malicious use such as fraud, manipulation, cyber attacks, weapon development, disinformation and surveillance.
risk.hci	Discussion of risks related to overreliance and loss of autonomy
risk.socioeconomic	Discussion of socioeconomic risks related to labour market, inequality, power concentration and environmental harms.
risk.safety	Discussion of risks related to misalignment, unpredictability, loss of control and ethical issues regarding sentient AI systems.

coping.emotion.cognitive_reappraisal	Instances where participants reinterpret AI-related risks in a way that makes them feel more manageable, such as reframing the risk as personally assessable, comparable to familiar risks, or limited by enduring human values such as creativity, judgement, or meaning.
coping.emotion.deliberate_optimism	Instances where participants consciously focus on positive possibilities, benefits, opportunities, or constructive uses of AI in order to reduce worry, maintain hope, or reassure themselves or others.
coping.emotion.fatalistic_reasoning	Instances where participants frame AI-related risks as inevitable, unavoidable, or beyond meaningful control, thereby reducing the perceived need for active

	problem-focused coping and managing emotional tension through acceptance.
coping.emotion.neutral_tool_reframing	Instances where participants describe AI as a neutral tool whose effects depend on human use, shifting perceived threat away from the technology itself and toward users, incentives, or surrounding systems.
coping.emotion.avoidance	Instances where participants reduce exposure to AI-related threat cues, such as avoiding AI-related media or discussions, in order to lower anxiety or emotional discomfort rather than directly addressing the underlying risk.
coping.problem.design.alternative_architectures	Proposals to redesign AI systems at a structural level to reduce risks, including new model types, training approaches, or domain limitations.
coping.problem.design.behavioural_friction	Design interventions that intentionally slow down, interrupt, or reshape user interaction patterns to reduce harmful or excessive use.
coping.problem.design.data_governance	Strategies focused on managing data access, privacy, and data quality to reduce AI-related risks.
coping.problem.design.technical_safeguards	Built-in technical protections aimed at preventing harmful outputs or misuse of AI systems.
coping.problem.design.transparency_features	Proposals to increase user awareness of how AI systems operate by making algorithmic processes visible or understandable within the interface.
coping.problem.design.use_ai_for_safety	Using AI systems themselves to detect, prevent, or mitigate risks caused by AI or other technologies.
coping.problem.economic_system_change	Proposals to change or reform economic systems in order to mitigate AI-related risks.
coping.problem.educate.critical_evaluation	Learning how to assess, verify, and question AI outputs (e.g., checking sources, recognizing hallucinations, not overtrusting).
coping.problem.educate.ethical_use	Learning norms or guidelines for responsible and morally appropriate AI use.
coping.problem.educate.practical_usage	Developing skills for effectively interacting with AI systems (e.g., prompt engineering, appropriate task allocation).

coping.problem.educate.technical_understanding	Seeking to reduce risk by understanding how AI systems function at a conceptual or technical level (e.g., training, hallucinations, statistical nature).
coping.problem.global_regulation_required	Statements emphasising that AI regulation must occur at a global or international level to be effective.
coping.problem.personal_research	Engagement in academic or research activities aimed at understanding, mitigating, or solving AI-related risks.
coping.problem.regulate_ai_development	Proposals or support for regulating how AI systems are developed, including through laws, policies, or institutional governance
coping.problem.regulate_ai_use	Proposals to regulate or restrict how AI is used by individuals or organisations.
coping.problem.responsible_use.avoid_in_education	Restricting or avoiding AI use in educational contexts to preserve learning and skill development.
coping.problem.responsible_use.calibrated_trust	Maintaining a balanced view of AI reliability, acknowledging both strengths and limitations.
coping.problem.responsible_use.critical_engagement	Maintaining an active, reflective, and questioning stance toward AI outputs and interaction.
coping.problem.responsible_use.domain_competence_required	Using AI only in areas where the user already has sufficient domain knowledge to evaluate outputs.
coping.problem.responsible_use.efficient_prompting	Optimising prompts to reduce unnecessary computational cost or resource use.
coping.problem.responsible_use.hil	Ensuring that a human reviews, controls, and / or takes responsibility rather than delegating autonomy to AI.
coping.problem.responsible_use.output_verification	Actively checking, comparing, or validating AI outputs using additional sources or methods.
coping.problem.responsible_use.selective_use	Restricting or avoiding AI use in specific contexts, especially high-risk or inappropriate domains.
coping.problem.responsible_use.thoughtful_u	Using AI deliberately with clear purpose, rather than indiscriminately or out of convenience.

se	
coping.problem.select_trusted_providers	Choosing AI tools or services based on perceived trustworthiness, transparency, or data handling practices.
coping.problem.shape_career	Intentional decisions or aspirations to choose career paths that allow the individual to mitigate, regulate, or responsibly influence AI development or use.
coping.problem.skill_differentiation	Focusing on developing skills or expertise in areas where AI is perceived to be less capable.
coping.problem.slow_down_development	Advocacy for reducing the pace of AI development to allow for reflection, evaluation, and safer implementation.
coping.problem.source_diversification	Actively seeking out diverse information sources to counteract algorithmic bias, filter bubbles, or misinformation.
coping.problem.strengthen_personal_profile	Actively building a stronger professional profile to remain competitive in an AI-influenced labour market.
coping.problem.technical_mitigation	Developing or proposing technical solutions aimed at directly reducing AI-related risks.

responsibility.self	Discussion of participant's own responsibility in mitigating an AI risk
responsibility.developer	Discussion of developer responsibility in mitigating an AI risk
responsibility.AI_companies	Discussion of AI companies' responsibility in mitigating AI risk.
responsibility.government/authorities	Discussion of government and authority responsibility in mitigating an AI risk.
responsibility.individuals	Discussion of personal responsibility of individuals in general, in mitigating an AI risk
scope.self	The subject of concern when discussing the risk is the participant.
scope.close_relations	The subjects of concern when discussing the risk are close relations to the participant such as family, friends or coursemates.
scope.other_groups	The subjects of concern when discussing the risk are

	groups not directly related to the participant.
scope.society	The subject of concern when discussing the risk is society as a whole.
coping.actor.self	A coping behaviour is discussed as being acted upon by the participant.
coping.actor.developer	A coping behaviour is discussed as being acted upon by developers.
coping.actor.government/authorities	A coping behaviour is discussed as being acted upon by the government or authorities.
coping.actor.other	A coping behaviour is discussed as being acted upon by other individuals.
ai_conception	Describing what an AI is or what capabilities AI systems do / can possess.
ai_conception.autonomous	Describing AI as an autonomous agent.
ai_conception.neutral_tool	Describing AI as a neutral tool.
probed	Topic introduced after interviewer prompt
not.probed	Topic introduced spontaneously

Appendix D - Recruitment letter

Invitation to Participate in an Interview Study on Risks Associated with AI

AI is developing rapidly and raises many questions about its consequences for both individuals and society at large. I am currently conducting my bachelor's thesis project in Computer Science at Mid Sweden University and I am seeking participants for an interview study on how students in IT-related programmes perceive potential risks associated with artificial intelligence (AI).

The purpose of the study is to better understand how these students reason about different types of AI-related risks, which risks are perceived as most significant, and how students think about possible ways to address such challenges. Students in these programmes are of particular interest, as they may go on to work closely with AI systems or contribute to their development.

The study is interested in your personal thoughts and reflections. You are being invited because you are a student in a program that meets the study's selection criteria.

How is the study conducted?

The study is carried out as a semi-structured interview where we discuss your thoughts on potential risks related to AI. The interview will take place via Teams or Zoom, using both audio and video. We will schedule a time that suits you, and you can participate from any private location with a stable internet connection. The interview is expected to take approximately 30–60 minutes. It will be recorded and later transcribed using an AI tool.

Participation is voluntary

Your participation in the study is voluntary, and you may withdraw at any time. If you choose not to participate or wish to withdraw, you do not need to provide a reason. If you wish to withdraw, please contact the person responsible for the study (see below).

Are you interested?

If you are interested in participating, you are warmly welcome to contact me via email to schedule an interview.

Kind regards,
Martin Granstad, magr2205@student.miun.se

Responsible for the study
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